

A stretching mechanism evokes mechano-electrical transduction in auditory chordotonal neurons

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eLife Assessment

This **valuable** study uses a sophisticated array of techniques to investigate the mechanisms through which the chordotonal receptors in the locust ear (Müller's organ) sense auditory signals. Ultrastructural reconstruction of the sensory organ provides **convincing** evidence of the organization of the scolopodial structure that wraps the sensory neuron cilium. However, the recordings of sound-evoked motion and electrophysiological activity from the chordotonal sensory neurons provide **incomplete** evidence for the proposed axial stretch model of mechanotransduction.

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Abstract

Insects' sound and vibration detection including proprioception rely on the scolopidium—a mechanosensory unit enclosing the sensory cilium of chordotonal organ neurons. The cilium, enclosed by a scolopale cell, contains mechanosensitive ion channels with the ciliary tip embedded in a cap. Despite knowledge of the scolopodial structure in multiple insects, the mechanism by which mechanical force elicits transduction remains speculative. We examined scolopidia in the auditory Müller's organ of the desert locust and present a comprehensive three-dimensional (3D) ultrastructure of a scolopidium using Focused Ion Beam Scanning Electron Microscopy (FIB-SEM). Next, we characterised sound-evoked motions of Müller's organ and the scolopidium using Optical Coherence Tomography (OCT) and high-speed light microscopy. Finally, we measured transduction currents via patch clamp electrophysiology during mechanical stimulation of individual scolopidia. By combining ultrastructure, sound-evoked motions, and transduction current recordings, our finding suggests that the scolopidium is activated best by stretch along the ciliary axis.

Introduction

All animal ears share the same fundamental purpose of converting sound-evoked vibrations into electrical signals¹. In insects, this process, known as mechano-electrical transduction (MET), occurs in the distal ciliated endings of chordotonal organ neurons². The cilium is enclosed in an evolutionarily conserved multicellular structure, the scolopidium³, where the cilium extends into a receptor lymph cavity made by the scolopale cell⁴. The ciliary tip is embedded in an extracellular cap tightly fused to an attachment cell. This stereotypical structure of a scolopidium is found in all insect chordotonal organs and underlies sound and vibration detection and proprioception⁵. Thus, understanding how the scolopidium functions has widespread implications for insect behaviour.

While it is established that MET takes place within the cilium, the precise mechanism remains speculative⁵. MET channels open in response to mechanical stimulation leading to a depolarisation that triggers downstream spikes. In *Drosophila* chordotonal organs, putative transduction channels, No mechanoreceptor potential C (NompC) and Nanchung-Inactive (Nan-Iav), are localised either side of the ciliary dilation—an enlarged region where the axonemal microtubules are farther apart than the rest of the cilium. Specifically, NompC localises to the distal end of the cilium, while Nan-Iav is found more proximally, extending from the ciliary rootlet to the dilation^{6–9}. Early studies proposed two primary models for how these channels open: axial extension (stretching)^{10–12} or lateral compression (pinching) of the ciliary tip¹³. One prominent axial extension model posits that stretching deforms the ciliary dilation, generating shear forces within the ciliary membrane that open the MET channels¹⁰. In contrast, the lateral compression model is supported by a similar proposed mechanism in hair plate sensilla, where direct force application on the cap evokes a maximal receptor potential¹³. Alongside the two MET channel opening models—axial extension and lateral compression of the ciliary tip—several other working mechanisms have been proposed based on early ultrastructural and electrophysiological studies⁵. Notably, recent experiments on bush cricket's hearing organ, crista acustica, suggest that the cap may tilt relative to the ciliary rootlet upon sound stimulation^{14,15}. Such a tilting motion could activate the MET channels by stretching the ciliary membrane as the cilium bends, by asymmetrical compressing of the ciliary tip, or through a combination of both.

A detailed ultrastructure of the scolopidium is key to understand its function as a mechano-electrical transducer^{3,11,16–19}. The first ultrastructural studies of the scolopidium stem from more than 65 years ago^{3,19}, and only recently has three-dimensional (3D) reconstruction using Focused Ion Beam-Scanning Electron Microscopy (FIB-SEM), been employed to examine a miniaturised chordotonal organ, the Johnston's organ of parasitoid wasp *Megaphragma viggiani*²⁰. A high-resolution 3D reconstruction that focuses solely on the scolopidium using FIB-SEM could provide new insights into its ultrastructure, especially how each component interacts.

Complementing these structural insights, efforts to understand the sound-evoked scolopidial motion have yielded valuable, albeit indirect, inferences: Stephen and Bennet-Clark's pioneering stroboscopy of the locust Müller's organ suggested significant phase lags and anti-phasic motion across the organ, potentially inducing strain in auditory neurons and scolopidia²¹. More recently, Optical Coherence Tomography (OCT) based vibrometry of the bush cricket's crista acustica indicated complex, multi-modal tilting of the ciliary tip, including 3D elliptical motions^{14,15}. Although providing valuable insight, these observations lack the spatial resolution to directly resolve scolopidia. This limitation highlights the critical need for direct visualisation of its sound-evoked motion, and measurement of transduction currents during direct mechanical stimulation of a scolopidium.

To address these gaps, we investigated scolopidia within Müller's organ of adult desert locusts, *Schistocerca gregaria*, a model organism that uniquely enables recordings of transduction channel currents from auditory neurons in an adult insect⁹. The desert locust's tympanal ears are located on its first abdominal segment (**Fig. 1a-b**). The locust's hearing organ, Müller's organ, is directly attached to the inside of the tympanic membrane at an inward fold of the tympanum called the elevated process (**Fig. 1b-c**). Auditory neuron somata locate mostly in the ganglion, while their dendrites extend to distinct regions of the organ, forming three groups also distinguished by characteristic frequency ranges^{9,22} (**Fig. 1c**, **Extended Data Fig. 1a-c**). This study examines scolopidia from Group-III auditory neurons (**Fig. 1c-d**), which constitute most auditory neurons (~48 out of ~80 neurons) in Müller's organ²³ and are tuned to 2.5-4 kHz¹, the frequency ranges associated with wing-beat noise²⁴.

Here, we combined FIB-SEM to resolve the 3D ultrastructure of a scolopidium, OCT and high-speed microscopy to examine sound-evoked motion at both the organ and individual scolopidium levels, and direct mechanical stimulation of the scolopale cap, where the ciliary tip is anchored, whilst simultaneously recording transduction currents. Understanding the structural and functional dynamics of scolopidia in insect auditory transduction is not only key to deciphering insect hearing but offers insights into basic principles of ciliary mechanosensation across insects.

Results

Three-dimensional (3D) ultrastructure of a scolopidium

To reconstruct a 3D ultrastructure of a scolopidium, we performed Focused Ion Beam-Scanning Electron Microscopy (FIB-SEM) at the interface between the scolopale cells and the attachment cells, where the scolopidia are located (**Fig. 2a-b**, see Supplementary Video 1) (N = 2, n = 3). In this region, the scolopale cell forms a protrusion that encloses the scolopale sheath with approximately half of the sheath inserting deep into the attachment cell. The scolopale sheath, previously described as composed of 5-7 partially fused, actin-rich concentric rods, is contained within the scolopale cell^{3,25}. In early ultrastructural studies, the distal region of the scolopidium was thought to end with the extracellular porous cap^{3,10,11,17}; however, our 3D reconstruction reveals a thin layer of fibrous filaments, likely actin filaments based on phalloidin staining of a similar structure in *Drosophila* scolopidia²⁶, within the attachment cell at the interface between the attachment cell and the scolopale sheath (**Fig. 2c**, arrowhead). Upon examining series of longitudinal sections along the length of the scolopidium, this fibrous material covers the distal end of the ciliated dendrite and the cap like a cone-shaped 'lid' (**Fig. 2d**, arrowheads). Thus, we describe the ensheathed part of a scolopidium as a cartridge composed of the scolopale sheath and a scolopale lid (**Fig. 2l**).

The scolopale cartridge encloses a tightly sealed receptor lymph-filled scolopale space that contains the ciliated dendrite and the cap (**Fig. 2b**). Our 3D reconstruction reveal that the cap resembles a truncated cone, with a central passage for the cilium that is funnel-shaped and is distinctly narrow at its entrance (**Fig. 2e-f**). An early ultrastructural study of Müller's organ reported that the cilium terminates within the cap³; however, we find that the cilium instead extends through the central passage and terminates at an electron-dense region at the tip of the cartridge, in the three scolopidia we imaged (**Fig. 2g**, **Extended Data Fig. 1d**). Interestingly, one of our datasets shows a break of the cilium near its terminus, which could have led to the assumption that the cilium terminates in the porous cap (**Extended Data Fig. 1e**). The cilium's full extension to its insertion at tip of the cartridge is a prominent, previously undescribed feature with clear mechanical implications.

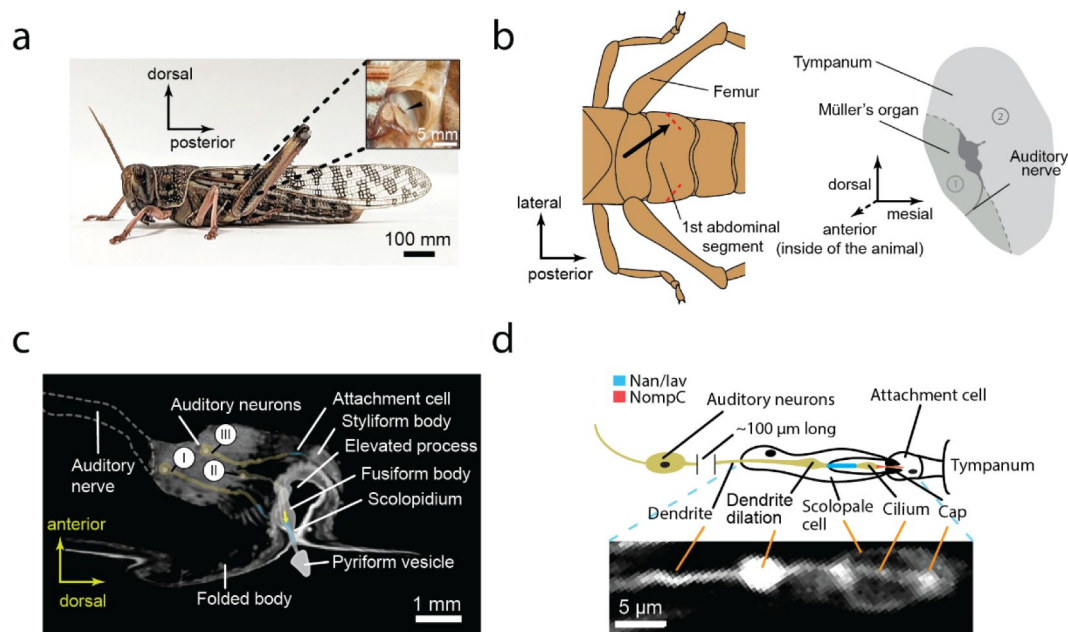


Figure 1.

The locust's ear, Müller's organ, and the scolodidium.

(a) Locust's ears are located on the first abdominal segment, hidden behind the wings. Each ear consists of a tympanic membrane with a hearing organ, Müller's organ, directly attached. External view of the tympanic membrane (inset) shows a dark ridge (arrowhead) where Müller's organ is located. The anatomical references used in this manuscript are defined based on the axes of the locust's body unless stated otherwise. **(b)** The surface of the tympanic membrane aligns orthogonally to the antero-posterior axis of the locust body at a slight oblique angle (left, red dashed lines). Illustration of the right ear viewed from inside the body (along the orientation of the black arrow on the left panel) shows Müller's organ located between the 'thick' (1) and 'thin' (2) regions of the tympanic membrane (right). The auditory nerve projects ventrally toward the ventral nerve cord. **(c)** Schematic of Müller's organ superimposed on an X-ray micro-CT scan slice shows three groups of auditory neurons. These neurons (yellow) extend their dendrites to distinct regions where the distal ciliated end of the dendrites insert into the attachment cells (blue). The auditory neurons of Müller's organ are classified based on their maximal frequency sensitivity. Group-I neurons respond to frequencies between 0.5 and 1.5 as well as 3.5 kHz (~20 cells). Group-II neurons are sensitive to higher frequencies of 10–25 kHz (~13 cells), while most auditory neurons, belong to Group-III and respond to mid-range frequencies of 2.5–4 kHz (~46 cells). The yellow arrow marks a typical location of a scolopidium at the insertion of the scolopale cell into the attachment cells. **(d)** Schematic of a scolopidium indicating locations of the mechano-electrical transduction (MET) channel candidates, NompC (red) and Nanchung-Inactive (blue), near the proximal and distal ends of the ciliary dilation (top). The bottom panel shows fluorescent micrograph of the corresponding region of the dendrite and the scolopidium. The auditory neuron is filled with Dylight-488 conjugated streptavidin-neurobiotin (bottom).

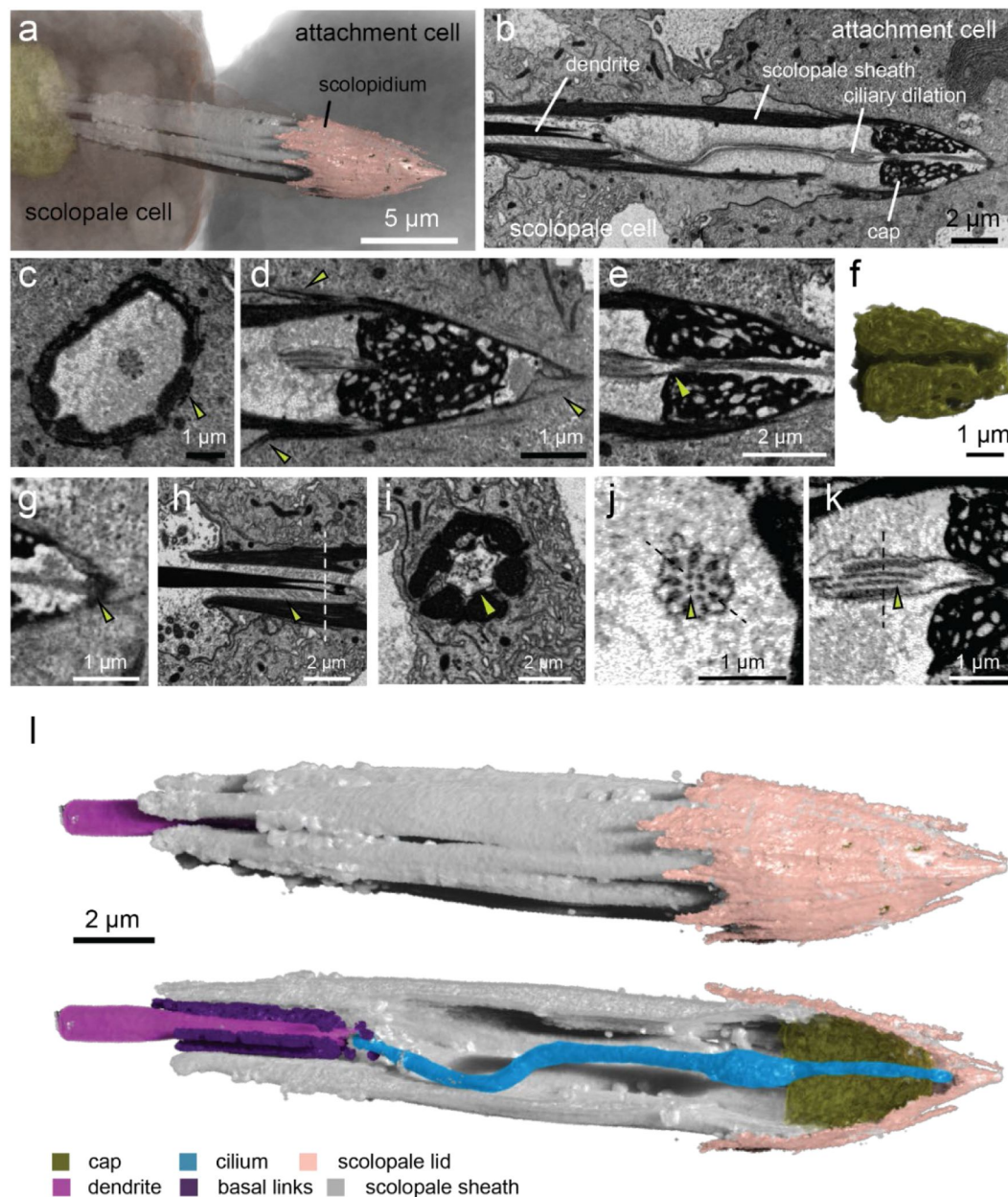


Figure 2.

Three-dimensional (3D) ultrastructure of a scolopidium.

(a) 3D reconstruction at the interface between a scolopale cell and an attachment cell using Focused Ion Beam-Scanning Electron Microscopy (FIB-SEM). (b) Longitudinal FIB-SEM slice reveals the dendrite and its ciliated distal end enclosed within the scolopale sheath; the cilium dilates near its insertion into the porous cap before it terminates in an electron-dense region at the border of the attachment cell. (c) Cross section at ciliary dilation shows fibrous filaments inside the membrane of the attachment cell (arrowhead). (d) Longitudinal section at the distal region of the scolopale sheath reveals that the fibrous filaments (arrowheads) observed in (c) form a cone-shaped lid covering the cap and the distal region of the ciliated dendrite. (e) Longitudinal section at the plane of the cilium shows the narrowing of the central passage, where the cilium enters (arrowhead). (f) 3D reconstruction of the cap material reveals a funnel-shaped central passage. (g) Termination of the cilium at the extreme end of the scolopale cartridge. (h) Longitudinal section at the scolopidium base reveals numerous oblique basal links between scolopale rods and the dendrite. (i) Cross section along the dashed line in (h) shows the basal links radiating from desmosome-like junctions. (j) Cross section at the ciliary dilation reveals four central fibrils (arrowheads) inside the 9-doublets axonemal structure. (k) Longitudinal section along the dashed line in (j) reveals two of the central fibrils along the length of the ciliary dilation. (l) 3D model of a complete scolopidium. The auditory neuron in the 3D model is omitted to reveal the oblique basal links.

At the proximal region of the ciliated dendrite, the cilium is anchored through rootlets. Here, numerous basal links connect the dendritic membrane to the base of the scolopale sheath and form an oblique angle of about 50° to the dendrite (**Fig. 2h**). The basal links show no apparent spacing and are found from the base of the scolopale sheath up to the proximal basal body, where the cilium inserts into the scolopale space. In a cross section orthogonal to the ciliary axis, the basal links appear to radiate towards the dendrite from desmosome-like junctions at the scolopale cell membrane opposing each scolopale rod (**Fig. 2i**). The basal links and their connections to the base of each scolopale rod have been previously shown^{10,11,27}, but not the oblique nature of the links.

Bending occurs consistently at a particular region within the proximal cilium in all our datasets, a feature previously observed in other chordotonal organs^{5,12,28}; however, we (and other studies) cannot rule out fixation artifacts being the cause of the bending due to preparation for electron microscopy. In contrast, the distal cilium remains straight (**Fig. 2b**, **l**). In the ciliary dilation, where the 9×2+0 axonemal microtubules are farther apart, we found 3–4 central fibrils within the restricted length of the dilation but nowhere else (**Fig. 2j–k**, **Extended Data Fig. 1f–g**). While these central fibrils have been mentioned by early ultrastructural studies^{3,10} and observed in crista acustica of the bush cricket, *Mecopoda elongata*¹⁴, their precise nature remains unclear.

In summary, our high-resolution 3D ultrastructural analysis has redefined the ensheathed part of the scolopidium as a ‘cartridge’ composed of the scolopale sheath and scolopale lid (**Fig. 2l**). This analysis also revealed the full extent of the cilium (**Fig. 2g**), the oblique nature of the basal links (**Fig. 2h**), and the central fibrils in the dilation (**Fig. 2j–k**). This refined anatomical understanding is crucial for interpreting how forces are transmitted within this mechanosensory unit. Next, we therefore examined sound-evoked motion at the level of Müller’s organ and individual scolopidia to determine how these intricate structures are mechanically stimulated.

Sound-evoked motion of Müller’s organ is tuned differently across its axes

The tympanic membrane vibrates in response to sound pressure changes in the air^{29–31}. This sound-evoked vibration is then transferred to Müller’s organ and mechanically activates the scolopidia inside^{22,29}. The resulting motion of the organ is complex and frequency-selective, arising from both the tonotopic traveling wave across the tympanic membrane and the organ’s attachment to mechanically heterogeneous regions on the tympanum^{21,29,31}. Ultimately, however, only the relative motion within the organ can produce forces that serve as the adequate stimulus for auditory transduction.

To identify the source of this relative motion, we measured the sound-evoked vibrations of Müller’s organ along two primary axes (**Fig. 3a**): (i) antero-posterior (motion orthogonal to the tympanic surface, **Fig. 3a-top panel**) and (ii) meso-lateral (motion tangential to the tympanic surface, **Fig. 3a-bottom panel**). The OCT imaging system generated sectional images across the two planes (**Fig. 3b**) and resolved most of the organ, apart from the region of Group-I scolopidia above the folded body, possibly due to low reflectivity (**Fig. 3b**). Notably, we could not resolve the Group-III scolopidia along the ventro-dorsal axis—which runs parallel to the dendrite—as the OCT beam was obstructed by either the cuticle or the elevated process.

We first examined the frequency response of Müller’s organ relative to sound stimulation in response to 90-dB-SPL tones (**Fig. 3c–d**). The vibration was axis-dependent: along the antero-posterior axis, the amplitudes peaked at 3.9 kHz, 9 ± 5 dB re 1 nm ($N = 3$, $E = 3$), while the meso-lateral axis showed a distinct peak at 2.4 kHz, 14 ± 5 dB re 1 nm ($N = 7$, $E = 12$) (**Fig. 3c**). There is also relatively large vibration at 1 kHz in both axes, antero-posterior: 14 ± 10 dB re 1 nm ($N = 3$, $E =$

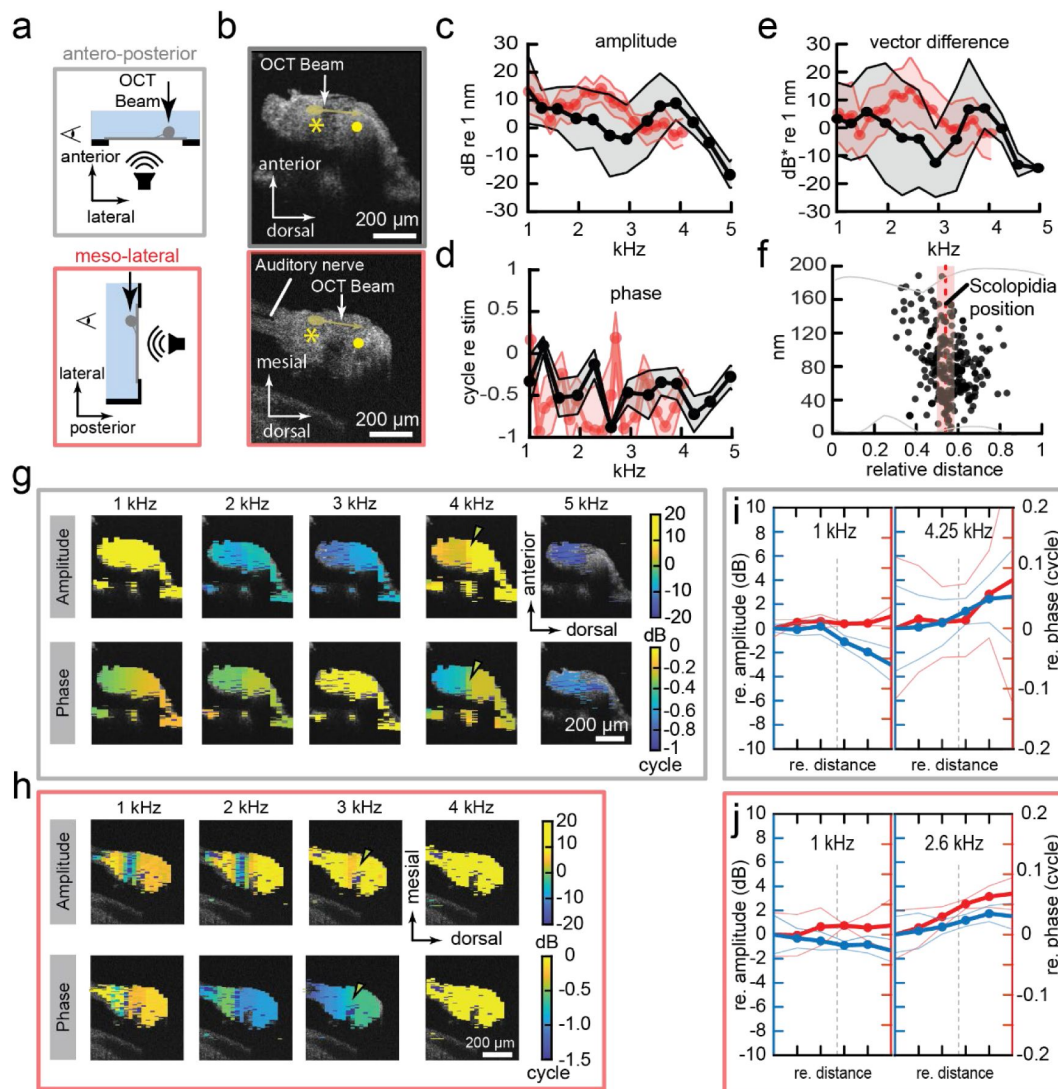


Figure 3.

Sound-evoked motion of Müller's organ.

(a) Vibrometry measurements were performed using Optical Coherence Tomography (OCT) along the antero-posterior (top) or meso-lateral (bottom) axes of Müller's organ upon 90 dB SPL sound stimulation, with the external side of the tympanic membrane facing the loudspeaker. Vertical arrows indicate the direction of the OCT beams, and the axes of vibration measured. The eyes represent the observation angles, where 2D OCT sections in **(b)** were reconstructed. **(b)** OCT sections along the two planes were reconstructed from OCT depth profiles. The symbols * and • indicate the maximal widths of Müller's organ at the ganglion and the styliform body, the two important landmarks used in our analysis. Yellow arrows indicate typical orientation of Group-III dendrites. **(c)** Ensemble amplitude and **(d)** phase responses to stimulated frequencies along the antero-posterior (black, $N = 3$, $E = 3$) and meso-lateral (red, $N = 7$, $E = 12$) axes. **(e)** Vector difference as a function of stimulated frequency, computed by comparing vibration at the position • relative to *, as shown in **(b)**, along the antero-posterior and meso-lateral axes. **(f)** Normalised positions of Group-III scolopidia between the ganglion and the styliform body ($N = 16$, $E = 16$, $n = 235$). The grey line indicates the outline of Müller's organ. **(g-h)** Vibration maps as a function of stimulated frequency along **(g)** the antero-posterior and **(h)** meso-lateral axes. Arrowheads in **(g-h)** indicate the area of abrupt transitions in both the phase and amplitudes. **(i-j)** Comparison between relative amplitude (blue) and phase (red) responses from the position * to • along **(i)** the antero-posterior ($N = 3$, $E = 3$) and **(j)** meso-lateral ($N = 7$, $E = 12$) axes. The relative amplitudes are selected from 1 kHz (minimal relative amplitude) and at the frequencies where the relative amplitude is largest in each axis, 4.25 kHz and 2.6 kHz, respectively. The relative distance is normalised with * as 0 and • as 1, as shown in **(b)**. The grey dashed line represents the average Group-III scolopidia position obtained from **(f)**. All vibration amplitudes are shown in dB relative to a 1-nm reference vibration. For additional data, see **Extended Data Fig. 2** and **Source Data Fig. 3**.

3) and meso-lateral: 13 ± 8 dB re 1 nm (N = 7, E = 12). The organ exhibited large phase lag relative to the sound stimulus in both axes (**Fig. 3d**). Vibration at 1-2 kHz and 3.75 kHz correspond to known resonance modes of the entire tympanum and Müller's organ region²⁹.

To assess the relative motion across the organ, we further computed vector difference, a measure of relative motion that accounts for differences in both vibration amplitude and phase (see Methods), at the styliform body (●, **Fig 3b**) relative to the ganglion (*, **Fig 3b**), the region where Group-III scolopidia are located. Quantification of relative motion and vector difference are key to understand where forces that ultimately open MET channels are generated. Vector difference revealed two prominent peaks at 3.9 kHz along the antero-posterior axis, 7 ± 6 dB (N = 3, E = 3), and 2.4 kHz along the meso-lateral axis, 14 ± 11 dB (N = 7, E = 12). The small vector difference at 1.25 kHz in both axes, antero-posterior: 3 ± 11 dB (N = 3, E = 3) and meso-lateral 3 ± 10 dB (N = 7, E = 12), indicates no or very small relative motion in the region of Group-III scolopidia at this frequency (**Fig. 3e**). Next, we mapped these complex, frequency-dependent vibration patterns onto the locations of Group-III scolopidia.

Relative motion at the site of Group-III scolopidia

We mapped Group-III scolopidia by filling their auditory neurons with the neural tracer neurobiotin, delivered through back-fill of the auditory nerve, then staining with fluorescently-conjugated streptavidin that bound to the neurobiotin³². Since ciliated dendrites within the scolopale sheaths were not consistently labelled, we used their dendrite dilations as reliable proxies (**Fig. 1d**). The position of each dendrite dilation was manually measured relative to the ganglion and the styliform body (**Extended Data Fig. 2a-c**). These dilations locate at a known distance, 11.1 ± 3.7 μ m (N = 20, n = 46), from the ciliary rootlet (**Extended Data Fig. 2d**). Ensemble normalised positions of the dendrite dilations revealed that Group-III scolopidia are clustered at a relative distance $54 \pm 4\%$ (N = 16, E = 16, n = 253) between the ganglion and the styliform body (**Fig. 3f**).

Having established the location of the Group-III scolopidia, we analysed the 2D vibration profiles to determine if this location corresponds to a region of high relative motion (**Fig. 3g-h**). Indeed, we observed that while motion was largely homogeneous at low (1 kHz) and high (5 kHz) frequencies, a distinct spatial transition emerged at the intermediate frequencies. Along the antero-posterior axis, an abrupt change in both amplitude and phase appeared from 2 kHz upwards, with the largest transition occurring at ~ 4 kHz (**Fig. 3g** arrowhead, **Extended Data Fig. 2e-f**). A similar sharp transition was observed in the meso-lateral axis, though its peak occurred at a lower frequency of ~ 3 kHz (**Fig. 3h**, **Extended Data Fig. 2g-h**). Interestingly, upon examining the vibration profile (**Extended Data Fig. 2e, g**), the source of vibration appeared to shift with frequency. At lower frequencies (1-2 kHz), the ganglion side of the organ vibrated more strongly (~ 2.5 -4 dB) than the styliform body, while at higher frequencies the vibration was larger at the styliform body and the tympanum (**Extended Data Fig. 2e-h**). Crucially, upon superimposing the normalised position of Group-III scolopidia on the frequency response for both axes, the onset of this abrupt transition in amplitude and phase co-localizes precisely with the average position of the Group-III scolopidia (**Fig. 3i, j** and **Extended Data Fig. 2i**). To understand how this complex relative motion activates the scolopidium, we recorded the sound-evoked motion of individual scolopidia with high-speed light microscopy.

Elliptical motion and stretching of a scolopidium along the ciliary axis

To examine sound-evoked motion at the single-scolopidium level, we captured its high-speed motion during sound stimulation (Supplementary Video 2) and achieved an average acquisition rate of $8,602 \pm 289$ frames per second (N = 12, n = 48). Upon initial image analysis, we found two prominent motion components (**Fig. 4a-b**): (i) stretching of the scolopidium (**Fig. 4a**, $t = 0.12$ and 0.24 ms) followed by (ii) pulling motion of the scolopale sheath along the ciliary axis towards

the attachment cell (**Fig. 4a**, [t = 0.36 ms](#)). The stretching appeared to occur concurrently with a shearing motion. This shear was most apparent with deformation of cell membranes at the cap and the scolopale base (**Fig. 4a**, yellow lines). To quantify these complex scolopidium motions, we tracked the positions of two clearly visible landmarks: the base of the scolopale sheath, or ‘the scolopale base’ and the extracellular cap (**Fig. 4a**).

First, we analysed the raw amplitude of the scolopale base and the cap. Upon sound stimulation, the entire scolopidium underwent elliptical motion (**Fig. 4c**). Notably, the motion of the scolopale base was generally more circular, while that of the cap was more elliptical, particularly between 1.5-3 kHz. At 1 kHz; however, the scolopidium showed a strong elliptical motion oblique to the ciliary axis. When stimulated at 3 kHz, the elliptical motion of the cap aligned with the ciliary axis and was most pronounced.

To investigate the principal axis of the sound-evoked motion, we decomposed the motion of both the cap and the scolopale base into components along the ciliary axis, d_x (**Fig. 4d**) and orthogonal to it, d_y (**Fig. 4e**). For both landmarks, motion along the ciliary axis was strongest, with a prominent peak at 3 kHz, base d_x : 66 ± 55 nm ($N = 6$, $n = 24$) and cap d_y : 197 ± 64 nm ($N = 6$, $n = 24$), (**Fig. 4d**, **Extended Data Fig. 3a**, **c**). In contrast, orthogonal motion was minimal, with only a shallow peak at the same frequency, base d_x : 47 ± 26 nm ($N = 6$, $n = 24$) and cap d_y : 70 ± 44 nm ($N = 6$, $n = 24$), (**Fig. 4e**, **Extended Data Fig. 3b**, **d**). Consequently, the ratio of axial-to-orthogonal motion, d_x/d_y , is consistent with elliptical motion shown in **Fig. 4c**, where cap d_x/d_y peaked sharply at 3 kHz, base d_x/d_y : 1.29 ± 0.66 ($N = 6$, $n = 24$) and cap d_x/d_y : 2.72 ± 1.00 ($N = 6$, $n = 24$), confirming that axial motion is the dominant component of the frequency response (**Fig. 4f**).

Since the adequate stimulus for MET is expected to arise from relative motion within the scolopidium, we next computed (i) the relative distance between the cap and the scolopale base, Δd , and (ii) shear motion at the cap region quantified by tracking the upper and lower borders of the attachment cell upon sound stimulation. Indeed, the relative motion was also strongest and sharply peaked at 3 kHz, Δd : 74 ± 48 nm ($N = 6$, $n = 24$) (**Fig. 4g-h**). The shear motion of the cap region; however, produced a distinct but broad peak around 3 kHz, 215 ± 200 nm ($N = 6$, $n = 22$) (**Fig. 4i**).

Direct pulling of the scolopidium cap evokes maximal transduction current

High-speed light microscopy revealed that scolopidial motion during sound stimulation occurred primarily along the ciliary axis, with additional components orthogonal to it (**Fig. 4c**). To identify the principal axis through which the scolopidium is stimulated, we directly mechanically stimulated the scolopidium cap with a stimulating pipette while measuring transduction currents via whole-cell patch-clamp recordings of the corresponding Group-III auditory neuron somata (**Fig. 5a**; Supplementary Video 3). We delivered a series of step displacements in two directions: a ‘pull-push’ (horizontal) motion along the ciliary axis (**Fig. 5b**) and an ‘up-down’ (vertical) motion orthogonal to it (**Fig. 5c**). Here, we showed that stimuli in all four directions evoked sharp, transient inward currents at the onset of the step (**Fig. 5d-e**). The most sensitive responses were to pull and push stimulations at 1 nm (**Fig. 5d**), whereas upward and downward stimulations produced clearly noticeable transduction currents from 5 nm and above (**Fig. 5e**). The signal-to-noise ratio in single trials was low due to spontaneous discrete depolarisations found in chordotonal organs.^{7,9,33} Upon averaging the responses over the total number of locusts, pull-push: $N = 18$, $n = 18$ and up-down: $N = 11$, $n = 11$, the ensemble average transduction current revealed a graded, non-linear relationship between stimulus and current (**Fig. 5f-g**).

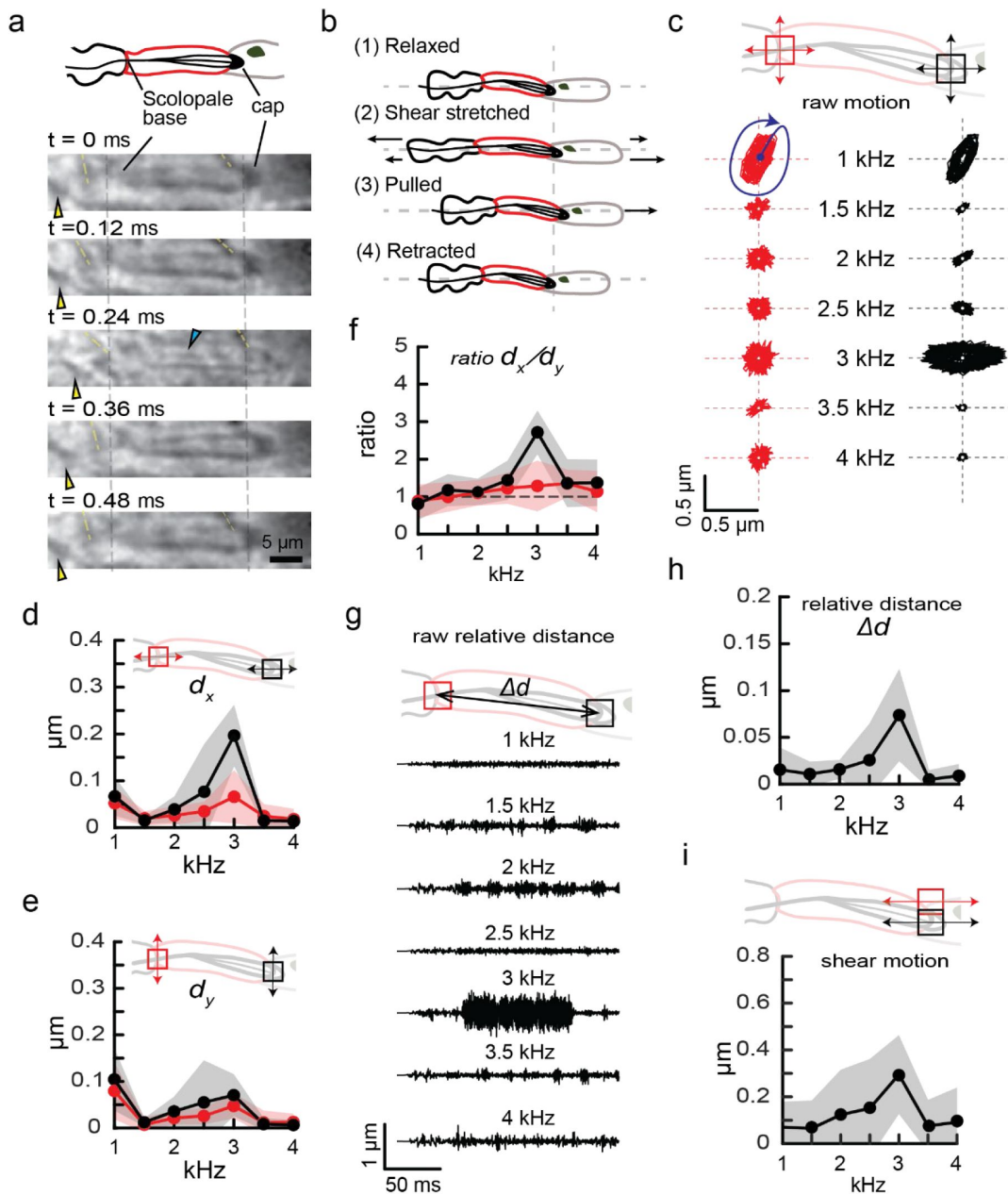


Figure 4.

Sound-evoked motion of the scolopidium.

(a) Schematic of a scolopidium showing the scolopale base and the cap that are clearly visible in light microscopy. (top). Montage over a period of one cycle of sound-evoked motion of a Group-III scolopidium stimulated with 3 kHz pure tone at 90 dB SPL (bottom). At $t = 0.24$ ms, the cyan arrowhead indicates the cilium when it is brought into focus. The yellow lines indicate the shear near the cap and the scolopale base. The yellow arrowheads follow the region with clear deformation near the scolopale base. (b) Summary of the sound-evoked motion of the scolopidium. (c) Motion of the scolopale base (red) and cap (black) regions along (d_x) and orthogonal (d_y) to the ciliary axis. (d) Root-mean-square (RMS) amplitude of the scolopale base (red, $N = 6$, $n = 22$) and the cap (black, $N = 6$, $n = 22$) regions along the ciliary axis (d_x). (e) RMS amplitude of the scolopale base (red, $N = 6$, $n = 22$) and the cap (black, $N = 6$, $n = 22$) regions along the orthogonal axis (d_y). (f) Ratio of the axial-orthogonal motion (d_x/d_y) of the scolopale base (red, $N = 6$, $n = 22$) and the cap (black, $N = 6$, $n = 22$). (g) Example of distance (Δd) between the scolopale base and the cap region as a function of time over each stimulated frequency. (h) Relative distance between the scolopale base and the cap ($N = 6$, $n = 24$). (i) Shear motion along the ciliary axis at the cap region ($N = 6$, $n = 22$). For additional data, see **Extended Data Fig. 3** and **Source Data Fig. 4**.

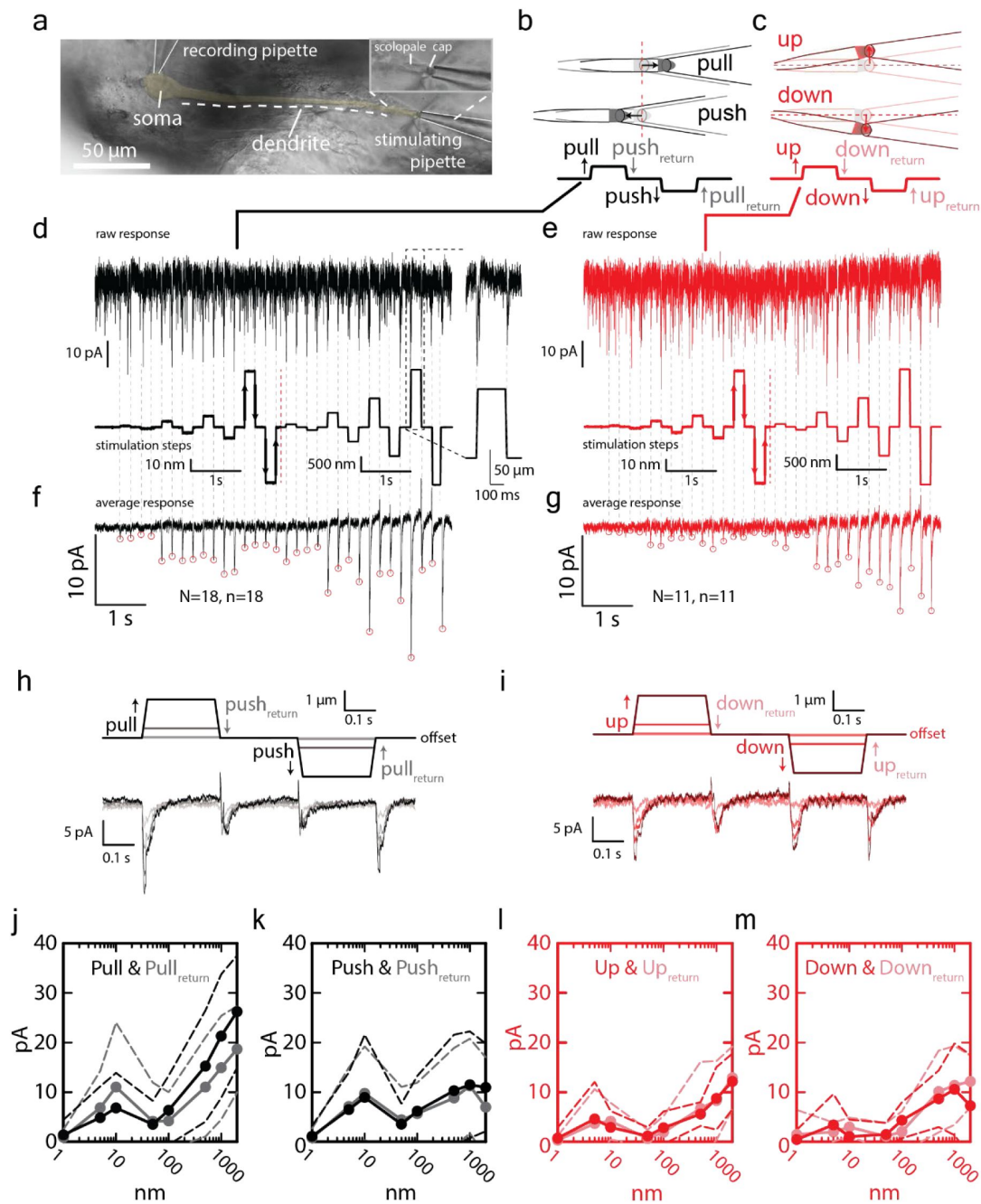


Figure 5.

Transduction current upon step displacements of a scolopidium cap.

(a) Focal-stacked light microscopy image from $\sim 10 \mu\text{m}$ apart showing the auditory neuron (yellow), a patch recording pipette, and a stimulating pipette. (b-c) Schematics of step-displacement protocols for (b) pull-push displacements along the ciliary axis ($N = 18$, $n = 18$) and (c) up-down displacement orthogonal to the ciliary axis ($N = 11$, $n = 11$). (d-e) Transduction currents (top trace) in response to (d) pull-push and (e) up-down step-displacements with increasing amplitudes (bottom trace). Step-displacement traces below and over 100 nm in (d-e) are shown in different scales. Arrows indicate the orientation of stimulation. (f-g) Ensemble average transduction currents in response to (f) pull-push and (g) up-down step displacements. The stimulation traces are shown in (d-e). (h-i) Examples of the transduction currents in response to (h) pull-push and (i) up-down step displacement of magnitudes 2, 50, 500, and 2000 nm . (j-m) Ensemble average transduction current plotted as a logarithmic function of the step-displacement, $I\text{-log}(X)$ curve, for (j) pull, (k) push, (l) up, and (m) downward displacements with their return to the offset counterparts. For additional data, see Source Data Fig. 3.

Next, we quantified transduction currents in response to pull-push step stimuli averaged across all recordings. There was a general increase in the magnitude of the transduction current with increased step displacement (**Fig. 5h**, **i**). Discrete depolarisations contributed to a large standard deviation of step-evoked transduction currents (**Fig. 5d**, **e**). There were distinct bi-phasic and logarithmic phases in the current-displacement relationship, I-log(X) curve, from 1-10 nm and 50-2000 nm in both pull-push (**Fig. 5j**, **k**) and up-down (**Fig. 5l**, **m**) displacements. The logarithmic relationship was determined by fitting linear regression models to the I-log(X) curves (see Source Data **Fig. 5**). The largest transduction currents were produced by pull along the ciliary axis. We quantified the difference at 2,000 nm displacement from the resting position between pull, 26.22 ± 10.99 pA ($N = 18$, $n = 18$), and push, 10.96 ± 8.59 pA ($N = 18$, $n = 18$) (Tukey's procedure, $p = 3.1e-05$, $t = -9.58$). Next, we found that the transduction current was larger for pull_{return}, 18.68 ± 8.43 pA ($N = 18$, $n = 18$) compared to push_{return}, 6.99 ± 9.75 pA ($N = 18$, $n = 18$) (Tukey's procedure, $p = 0.003$, $t = -4.72$). Additionally, the transduction current evoked by pull from the resting position was larger than displacement upward, 12.17 ± 5.37 pA ($N = 11$, $n = 11$) (Tukey's procedure, $p = 1.75e-03$, $t = -3.83$) or downward 7.28 ± 9.76 pA ($N = 11$, $n = 11$) (Tukey's procedure, $p = 5.10e-06$, $t = -4.53$). There was no difference between pull_{return} and up_{return} (Tukey's procedure, $p = 0.69$, $t = -1.91$), and between pull_{return} and down_{return} (Tukey's procedure, $p = 0.55$, $t = -2.24$).

Discussion

The scolopidium is the basic sensory unit of mechano-electrical transduction (MET) in chordotonal organ of insects^{3,10,12,13,17,27}. Addressing the fundamental question of how the scolopidium operates, we demonstrate here that axial stretch along the cilium is the adequate stimulus or the key mechanical input that activates mechano-electrical transduction. This conclusion is based on two key experiments. First, high-speed imaging revealed that elliptical sound-evoked scolopidium motion is dominated by stretch along the ciliary axis (**Fig. 4**). Second, pull of the scolopidium along the ciliary axis evoked the maximal MET current (**Fig. 5**).

Axial stretch is assumed to be the adequate stimulus in other chordotonal organs, based on either ultrastructural analysis or electrophysiological recordings, but this has never been experimentally confirmed with direct observation of the underlying mechanics^{10,12,34}. These early studies, performed before the molecular identity and precise location of MET channel candidates like NompC and Nan-Iav were known, led to at least ten proposed mechanisms^{5,35}. For example, deformation of the ciliary dilation^{10,12,34}, bending of the cilium^{12,34} or scolopale sheath^{25,36}, and contraction of the ciliary rootlet³⁶. Yet another hypothesis proposed that the cap that encloses the cilium could narrow during axial stretch and exert a compressive force to the ciliary membrane¹³. Our finding provides the first direct evidence that axial stretch is the adequate stimulus of this highly conserved mechanoreceptor found in all insects and some crustaceans³⁷.

How does stretch of the scolopidium lead to conformational change of MET channels within the cilium? Here, we propose four competing hypotheses (**Fig. 6**): (i) In the '*direct-stretch model*', axial tension is transmitted along the ciliary membrane, stretches the lipid bilayer and opens MET channels. (ii) In the '*stretch-compression model*', axial stretch narrows the central passage of the cap thus compressing the ciliary tip i.e. when a compressive material with a hollow tube is deformed along its axis, the tube narrows (like a 'Chinese finger trap'). (iii) In the '*stretch-deformation model*', the larger ciliary dilation is pulled against and into the narrow central passage of the cap (**Fig. 2e**, **f**). (iv) In the '*stretch-tilt model*', the cap pivots due to the shear component of the scolopidium stretching, this pivoting leads to either asymmetrical axial tension along the ciliary membrane or asymmetrical compression of the narrow central passage against the ciliary tip. In short, the proposed models differ in the primary force that opens the MET channels. The MET channels would be opened either through membrane tension in the direct-stretch model, or by compressive forces of the cap wall directly onto the membrane in the stretch-compression and

stretch-deformation models. However, in the stretch-tilt model the MET channels could be activated through either membrane tension as the cilium bends, lateral compression as it presses against the cap wall, or a combination of both.

While a complete gating model for insect MET remains elusive³⁸, key molecular insights have been provided by the cryo-EM structure of the putative channel NompC³⁹, as well as by Molecular Dynamics (MD) simulations^{40,41}, and electrophysiological recordings^{40,42,43}. Specific structures within NompC, such as its Ankyrin Repeat (AR)^{43,44} and Linker Helix (LH)⁴² domains, have been proposed to form a “gating spring”—a compliant element that transmits mechanical force into the channel’s conformational opening^{42,45}.

Currently, there is no direct evidence to support the direct-stretch model. However, there are lines of compelling evidence that support the stretch-compression, stretch-deformation, and stretch-tilt models. In particular, NompC expressed in *Drosophila* S2 transfected cells are more sensitive to membrane compression than suction⁴⁰. The other MET channel candidate Nan-Iav is not mechanosensitive when expressed in HEK293T cells⁸. The properties of both of these Transient Receptor Potential family ion channels may operate differently in the cilium compared to heterologous cell lines. Nanchung-Inactive is expressed below the ciliary dilation in the proximal ciliary segment. Therefore, if Nanchung-Inactive ion channels were mechanically sensitive *in vivo* they could only be opened by stretch as the stretch-compression, stretch-deformation and stretch-tilt models act only on the ciliary tip. Our OCT-based vibrometry reveals that axial stretch of the scolopidium arises from complex 3D motion of Müller’s organ²¹, a feature that maybe shared by other chordotonal organs^{14,46,47}. We demonstrate an abrupt transition in both phase and amplitude localised at the site of Group-III scolopidia, which would generate relative motion across the ciliary axis (**Fig. 3 g**, **h**). While motion along the antero-posterior axis is largely dominated by the tympanum’s movement at 3.5-4 kHz²⁹, strong vibrations along the meso-lateral axis at lower frequencies, 2.25-3 kHz (**Fig. 3 c**, **e**), may contribute to broad frequency sensitivity of the sound-evoked transduction current¹. In this experiment, we measured vibration along the two axes independently. To fully capture complex 3D motion of Müller’s organ, simultaneous measurement of vibration in both axes warrants further investigation, using either a two-OCT recording beam setup or a mirror to reflect the OCT beam onto an additional axis⁴⁸.

Finally, our high-resolution 3D ultrastructure provides the anatomical framework for how the scolopidium operates (**Fig. 2**). The presence of the scolopale ‘lid’ and the full extent of the cilium to its insertion at the scolopale cartridge tip indicate a firm, direct coupling between the attachment cell and the cilium, essential for a strong anchor at the cilium’s distal end. Furthermore, the distinct narrowing at the cilium’s entrance to the central passage may serve to compress or deform the cilium during axial stretch (**Fig. 2 e**, **f**). The numerous oblique basal links may not be directly involved in transduction but likely provide mechanical stability to the distal dendrite and ciliary rootlet (**Fig. 2 h**, **i**). The nature of the central fibrils within the ciliary dilation (**Fig. 2 j**, **k**) remains unknown but are also in bush cricket and mosquito auditory neuron cilia^{14,49}. Our OCT and high-speed camera measurements are from *ex vivo* preparations immersed in saline solution and our transduction current recordings are from enzymatically dissociated scolopidia. Despite these manipulations, the electrophysiological function of the auditory neurons appears physiologically normal when compared against preparations that more closely mimic *in vivo* conditions^{4,9,14,33,50}. In our previous work, we showed that, Group-III neurons retain their frequency tuning even when Müller’s organ and the internal surface of the tympanum is bathed in saline¹. Thus, we conclude that the electrophysiology of the auditory neurons and the mechanics of Müller’s organ and the scolopidia reported here mimic *in vivo* conditions.

In conclusion, our study provides a comprehensive, multi-scale analysis of MET in an insect auditory chordotonal neuron, using excised preparations of locust ears. We have shown that complex 3D motion of Müller’s organ produces localised relative motion precisely where the

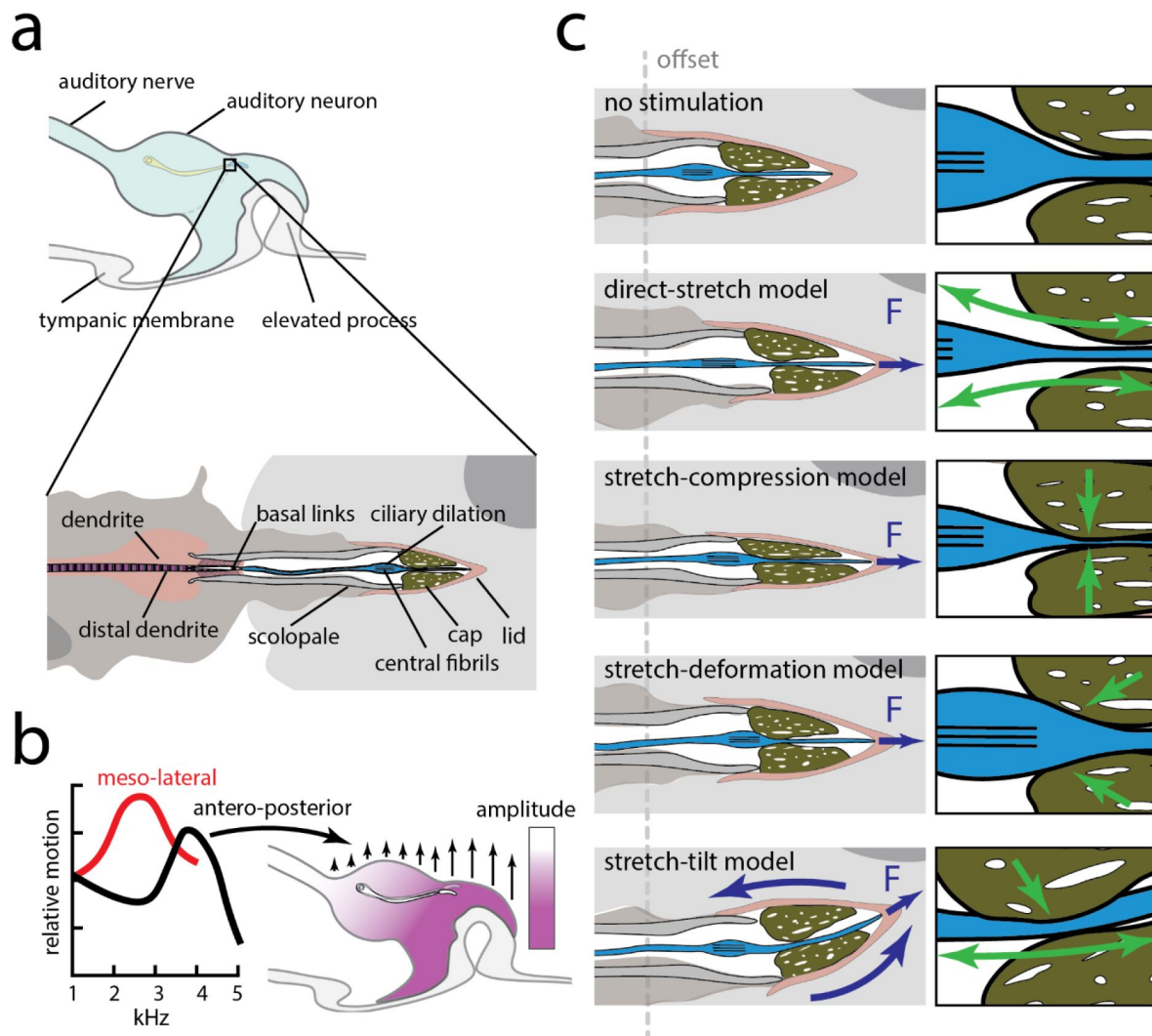


Figure 6.

Possible gating mechanisms of MET channels in insect chordotonal neurons.

(a) Key anatomical features of Müller's organ and the scolopidium. **(b)** Sound-evoked motion of Müller's organ generates axial stretch at the level of the scolopidium. **(c)** Possible gating mechanisms of the mechano-electrical transduction channels in insect chordotonal neurons (left): (i) direct-stretch model, (ii) stretch-compression model, (iii) stretch-deformation model, and (iv) stretch-tilt model. Purple arrows indicate forces exerting on the scolopidium. The right panel shows magnified views of the entry of the cap's central passage, showing forces (green arrows) exerting on the ciliary membrane.

auditory scolopidia are situated. This relative motion within the organ then drives an axial stretch of the scolopidium, which in turn evokes the mechano-electrical transduction current. As scolopidial structure is ubiquitous across insect classes, we find it probable that one of the four mechanisms of MET channel opening that we proposed here is largely conserved across insects.

Methods

Animals and sample preparation

Desert locusts, *Schistocerca gregaria*, were reared under crowded conditions with a 12:12-hour light/dark cycle and the ambient temperature at 36 °C (light) and 25 °C (dark). They were fed fresh wheat and bran. All experiments used adult male and female locusts 10-20 days after their final moult.

We used an excised locust ear preparation for all experiments, following a previously described dissection protocol⁹. In short, the ear was surgically removed and affixed to a custom-made 3D-printed plastic dish, positioning the tympanum over a 3 mm hole. For all measurements, the external side of the tympanum faced the sound source, while the internal side, including Müller's organ, was submerged in locust extracellular saline. The extracellular saline contained 185 mM NaCl, 10 mM KCl, 2 mM MgCl₂, 2 mM CaCl₂, 10 mM HEPES, 10 mM trehalose, and 10 mM glucose. The pH was adjusted to 7.2 using NaOH.

Sample preparation for FIB-SEM and 3D reconstruction of a scolopidium

Excised locust ears were fixed in a solution of 2.5% glutaraldehyde and 2% paraformaldehyde in 0.1 M sodium cacodylate (pH 7.2) with 4.1% sucrose. After washing, the samples were incubated in 1% osmium tetroxide with 1.5% potassium ferricyanide, followed by another wash and a 20-minute incubation in thiocarbonylhydrazide. Samples were then washed and sequentially incubated in 2% aqueous osmium tetroxide and 2% uranyl acetate, with washing steps in between. Following fixation and staining, the samples underwent serial dehydration in ethanol (30%, 50%, 70%, 90%, and 100%). The samples were then washed in propylene oxide before being embedded in a mixture of hard-formula TAAB 812 resin and propylene oxide. The resulting resin blocks were trimmed to the region of interest on an ultramicrotome (Leica UC7), which was verified by staining 100 nm-thick sections with methylene blue.

Serial acquisition was performed on a dual-beam FIB-SEM (Zeiss Crossbeam 500) equipped with a Gemini2 SEM column and a Ga⁺ source. Imaging parameters were set as follows: SEM extra high tension (EHT) at 1.6 kV with a 600 pA beam current, and FIB current at 700 pA at 30 kV. The voxel sizes of the acquired FIB-SEM stacks ranged from 9 to 15 nm. The 3D ultrastructure of the scolopidia were reconstructed from the FIB-SEM stacks using image processing software Dragonfly 3D.

Optical Coherent Tomography Vibrometry

We measured sound-evoked vibrations using a spectral domain Optical Coherence Tomography (OCT) system (Ganymede, Thorlabs). The field of view was 10 × 10 mm with a depth range of 3.5 mm. The OCT beam was directed from above, while the preparation was mounted either horizontally (with the sound source below, **Fig. 3a**) or vertically (with the sound source to the side, **Fig. 3b**). The excised ears were submerged in large amount of saline solution (~10 mm liquid high above the surface of the tympanic membrane) to minimise drifting over the time course of acquisition due to evaporation. The beam was manually aligned for each measurement axis to maximize the signal-to-noise ratio for that orientation. Sound stimuli, consisting of broadband multitone ‘zwuis’ complexes⁵¹, were generated using a Tucker-Davis Technologies

System III and delivered via a ScanSpeaker R2904/700 005 loudspeaker, placed at 32 cm from the preparation. The multitone ‘zwuis’ method allows for measurement of vibration amplitudes and phases from several frequencies simultaneously. The stimulus level used was 90 dB SPL; based on previous work, this level in a liquid-backed preparation corresponds to approximately 70 dB SPL in an air-backed tympanum.⁵²

OCT acquisitions were externally triggered and phase-locked to the acoustic stimulation at a sampling rate of 90 kHz. To characterize the organ’s complex 3D motion, vibrations were measured separately along two primary axes: antero-posterior (motion normal to the tympanic surface) and meso-lateral (tangential, side-to-side motion). The vibration magnitude is heavily damped but comparable to those submerged in a similar level of saline solution and measured using Laser Doppler Vibrometer (LDV).⁵² Vibration waveforms were derived from contiguous groups of three pixels per scan. All organ vibrations were measured relative 1 nm. The significance of phase-locking for each frequency component was assessed using the Rayleigh test ($p = 0.001$). All data acquisition and analysis were performed using a custom MATLAB toolbox (MathWorks, 2007) developed by Marcel van der Heijden.

The vector differences (**Fig. 3e**) were computed from the complex amplitudes of the two vibrations. Each vibration was represented as a complex number, $AA \cdot e^{i\phi}$, where AA is the vibration amplitude and ϕ is its phase relative to the stimulus. The vector difference represents the magnitude of the resulting complex number after subtracting the complex amplitude at the styliform body from that at the ganglion.

Localisation of Group-III scolopidia

To prepare the locusts for neural tracing, each insect was secured on its back in plasticine. A small section of the cuticle on the second and third thoracic segments was carefully cut and temporarily removed. After clearing away the tracheal air sacs to expose the metathoracic ganglia and the auditory nerve (nerve VI), the thoracic cavity was filled with a saline solution. The auditory nerve was then severed near the ganglion, and the cut end was positioned on the outer cuticle at the rear of the thorax. A small well was made around the exposed nerve ending using petroleum jelly and filled with a 5% Neurobiotin solution to allow the tracer to diffuse along the nerve. The well was sealed with more petroleum jelly, and the previously removed cuticle was placed back onto the original area and also sealed to prevent the area from drying out. The locusts were then incubated overnight at 4°C to facilitate the diffusion of Neurobiotin along the auditory neurons. After the incubation, the locust ears were dissected from the first abdominal segment. These ears were fixed in a 4% paraformaldehyde solution for 24 hours at 4°C. Subsequently, the fixed ears were washed three times in Phosphate Buffer Saline (PBS). To prepare for imaging, the ears were treated to make the cells permeable and to block non-specific binding. They were then incubated overnight with Dylight 488 streptavidin, which binds to the Neurobiotin to stain the auditory neurons, and DAPI, which stains the cell nuclei. Finally, after another series of washes in PBS, the ears were dehydrated using an ethanol series and cleared with Methyl salicylate. The stained neurons were then imaged using a confocal microscope.

The positions of dendrite dilations were then manually measured in Fiji.⁵³ The positions were recorded and saved as the region of interest (ROI) files and further processed in MATLAB R2004a (Mathworks). The position of each dendrite dilation was computed as a relative distance between the ganglion and the styliform body (**Extended Data Fig. 2**).

Tracking sound-evoked scolopidium motion

We recorded the sound-evoked motion of the scolopidium using a high-speed camera (KINETIK, Teledyne) mounted on an FN1 Nikon upright microscope using a 40X/0.8W WD = 3.5 mm (NIR Apo DIC N2) objective. Using an objective with a small working distance enabled us to minimise the saline volume, which reduced damping and maximised the amplitudes of scolopidial motion

within the detection range of our imaging system. We achieved an average acquisition rate of $8,602 \pm 289$ frame per second (24 movies). Sampling with an acquisition rate that did not achieve Nyquist sampling of its stimulated frequency were excluded from analysis. Sound stimulation was synchronised with imaging and delivered through a speaker (RS PRO 50 mm, RS PRO) via an analogue-to-digital converter (National Instruments PCIe 6323). The sound pressure level was calibrated by measuring the voltage output of an amplified microphone (Clio PRE-01 and Clio MIC-03, Audiomatica) in response to a 94 dB SPL 1 kHz tone from a sound level calibrator (Cal73, Brüel & Kjær). This calibration allowed us to determine the sound pressure level from the microphone's voltage output. The microphone was then positioned where the excised locust ear would be during experiments, and speaker voltages were adjusted to produce a flat response across the stimulated frequency range (1–5 kHz). An in-house MATLAB graphical user interface controlled the setup. In these experiments, we stained the excised locust ear preparation with 0.1% Janus Green B (201677-5G, Sigma Aldrich) to enhance the contrast of the scolopidium with its surrounding tissue.

The sub-pixel resolution motion of the scolopidium was tracked using Fiji/ImageJ Template Matching plugin (<https://sites.google.com/site/qingzongtseng/template-matching-ij-plugin>)⁵⁴. The tracking method is based on Normalised Correlation Coefficient and Gaussian peak fitting to determine the location of the correlation peak. The tracking achieved sub-pixel resolution of tens of nanometres. The coordinates output from the plugin were further analysed in MATLAB R2004a (Mathworks).

Whole-cell patch clamp and mechanical stimulation of a scolopidium

To expose Group-III auditory neurons for patch-clamp recordings, the fibrous tissue enclosing the medial–dorsal border of Müller's organ was digested by applying a solution of collagenase (0.5 mg/ml) and hyaluronidase (0.5 mg/ml; C5138 & H2126, Sigma-Aldrich) in extracellular saline. The enzymes were applied locally using a wide-bore (12 μ m) patch pipette, which was also used to gently remove the softened tissue via suction, exposing the somatic membrane of the neurons.

To prepare for mechanical stimulation, the cap of a targeted scolopidium was released from its attachment cell by puffing the same enzyme solution from the same wide-bore pipette. A new pipette filled with extracellular saline was then used to hold the cap via gentle, continuous mouth suction, which was renewed approximately every minute, and before each recording series, to ensure a secure connection. A series of push-pull (axial) and up-down (normal) step displacements (1, 5, 10, 50, 100, 500, 1000, and 2000 nm) were delivered to the stimulating pipette using a PatchStar micromanipulator (Scientifica).

Scolopidia were visualized with a custom upright microscope (Cerna® Mini, Thor Labs) equipped with a 40 $\times$ water-immersion objective (0.8 NA, 2 mm working distance; CFI APO NIR, Nikon). Patch-clamp electrodes (3–4 M Ω) were fabricated from borosilicate glass (GB150–8P, Science Products) using a vertical pipette puller (PC-100, Narishige). The intracellular (pipette) solution contained (in mM): 170 K-aspartate, 4 NaCl, 2 MgCl₂, 1 CaCl₂, 10 HEPES, 10 EGTA, and 20 tetraethylammonium (TEA). The extracellular saline contained (in mM): 185 NaCl, 10 KCl, 2 MgCl₂, 2 CaCl₂, 10 HEPES, 10 trehalose, 10 glucose, and 0.09 tetrodotoxin (TTX) citrate. To isolate and increase the magnitude of the transduction current, spikes were blocked with extracellular TTX and intracellular TEA. Electrophysiological signals were amplified, digitized, and recorded using an EPC 10 USB patch-clamp amplifier (HEKA Elektronik).

The data analysis was done in MATLAB R2004a (Mathworks). Peak detections of the ensemble average transduction current (**Fig. 5 f**, **g**) were done by detecting peak within a 30-ms windows at the onset of step-stimulation using Matlab built-in peak detection package 'findpeaks'.

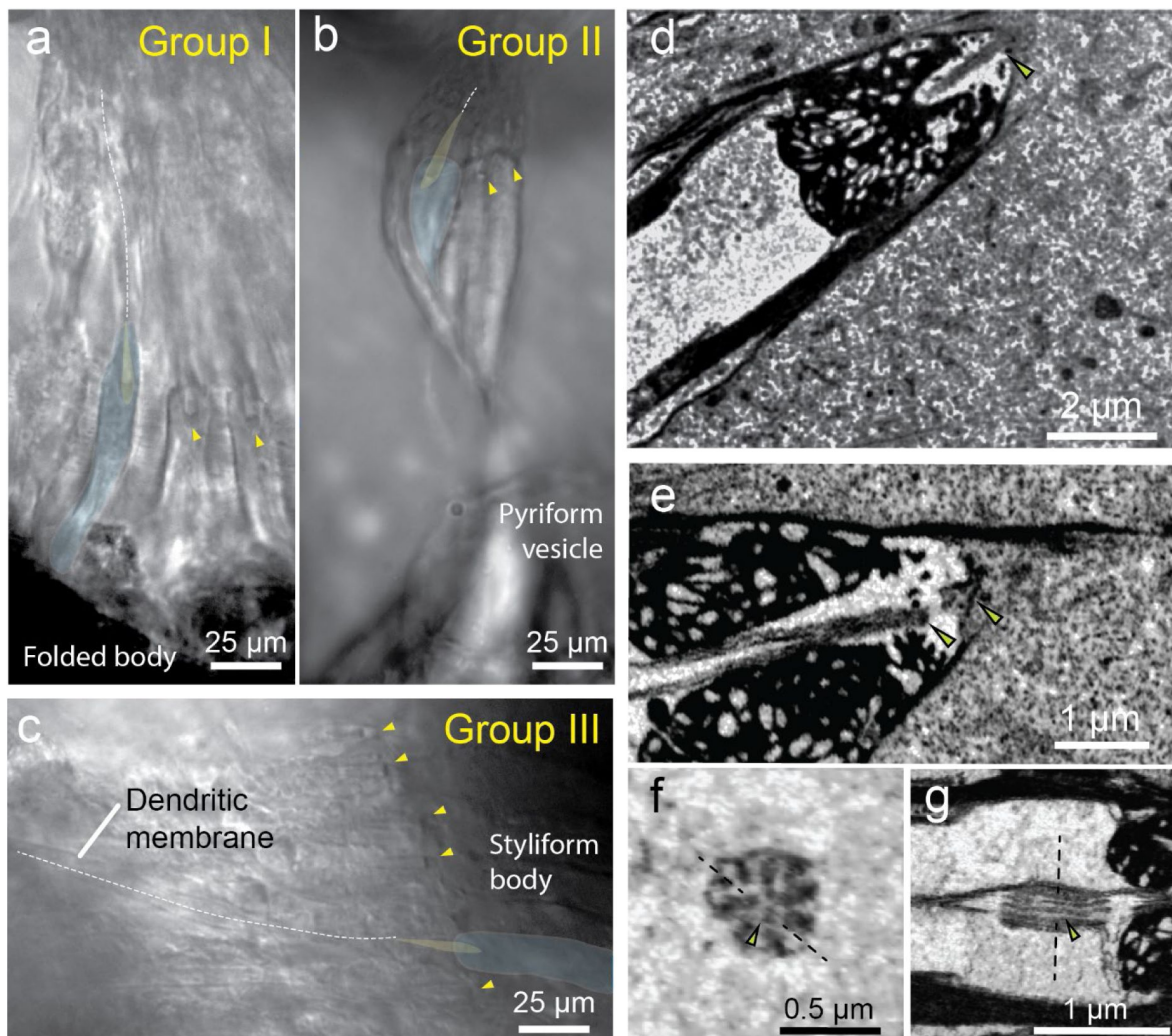
Statistical analysis

Descriptive data are presented as mean $\pm$ standard deviation. Biological replicates are denoted by N (number of locusts), E (number of ears), and n (number of cells). For the statistical analysis in **Fig. 5j-m** [↗](#), we fitted linear regression models to assess the correlation between the transduction current and the logarithmic step displacement from 50-2000 nm. To test for statistical differences, we performed a one-way ANOVA followed by Tukey's Honestly Significant Difference (HSD) post-hoc test for multiple comparisons. The reported p-values were obtained from the Tukey HSD procedure, while the t-statistics are from a Student's t-test. Further details on the statistical analysis are available in the source data.

Data availability

Raw data and MATLAB codes supporting plots in this manuscript can be found in the online repository Zenodo⁵⁵ [↗](#). Source data for each figure are provided with this paper as stated in the figure legends. Additional data can be requested from the corresponding author upon reasonable request.

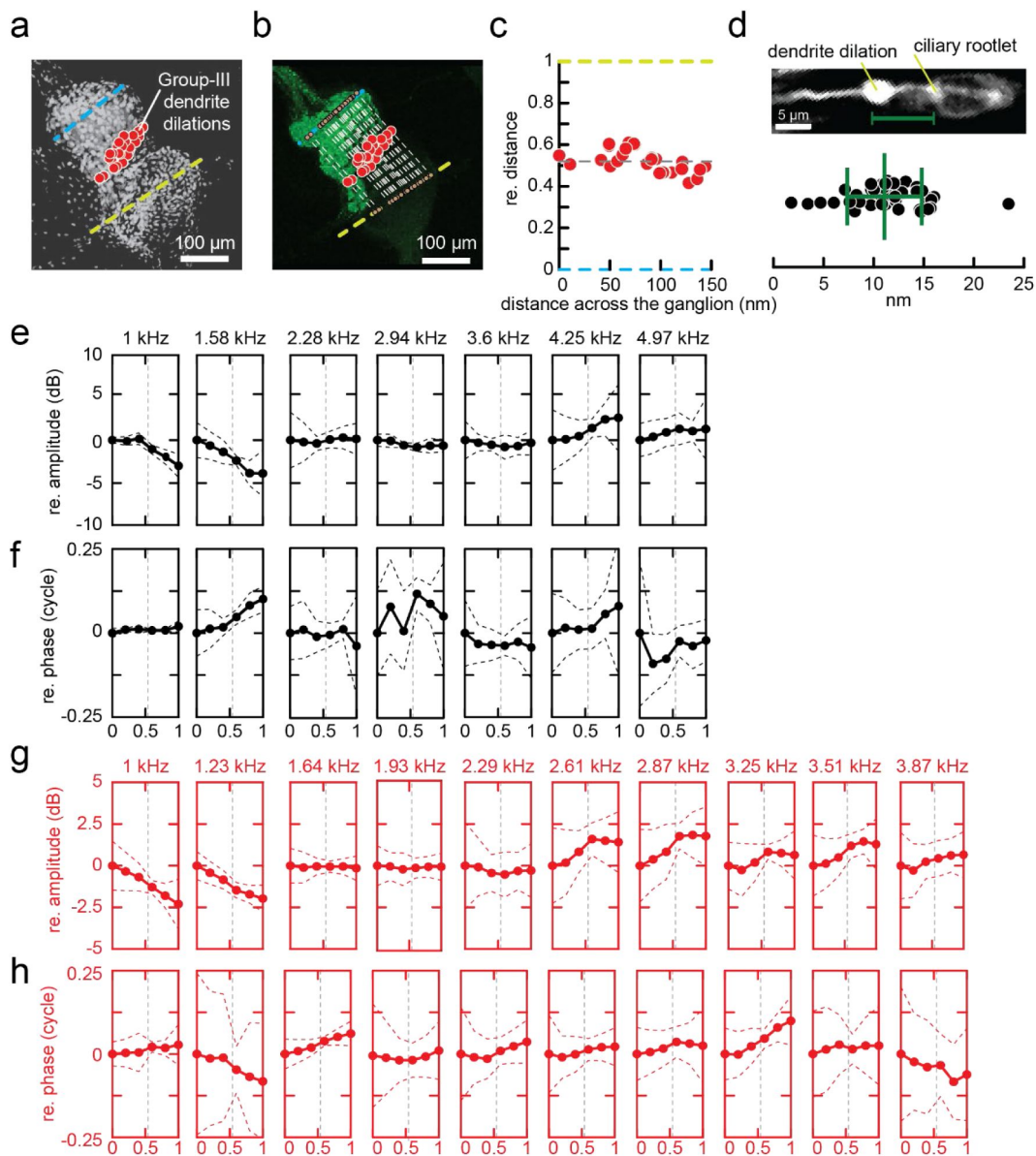
Extended data figures



Extended data figure 1.

Anatomy of the scolopidia.

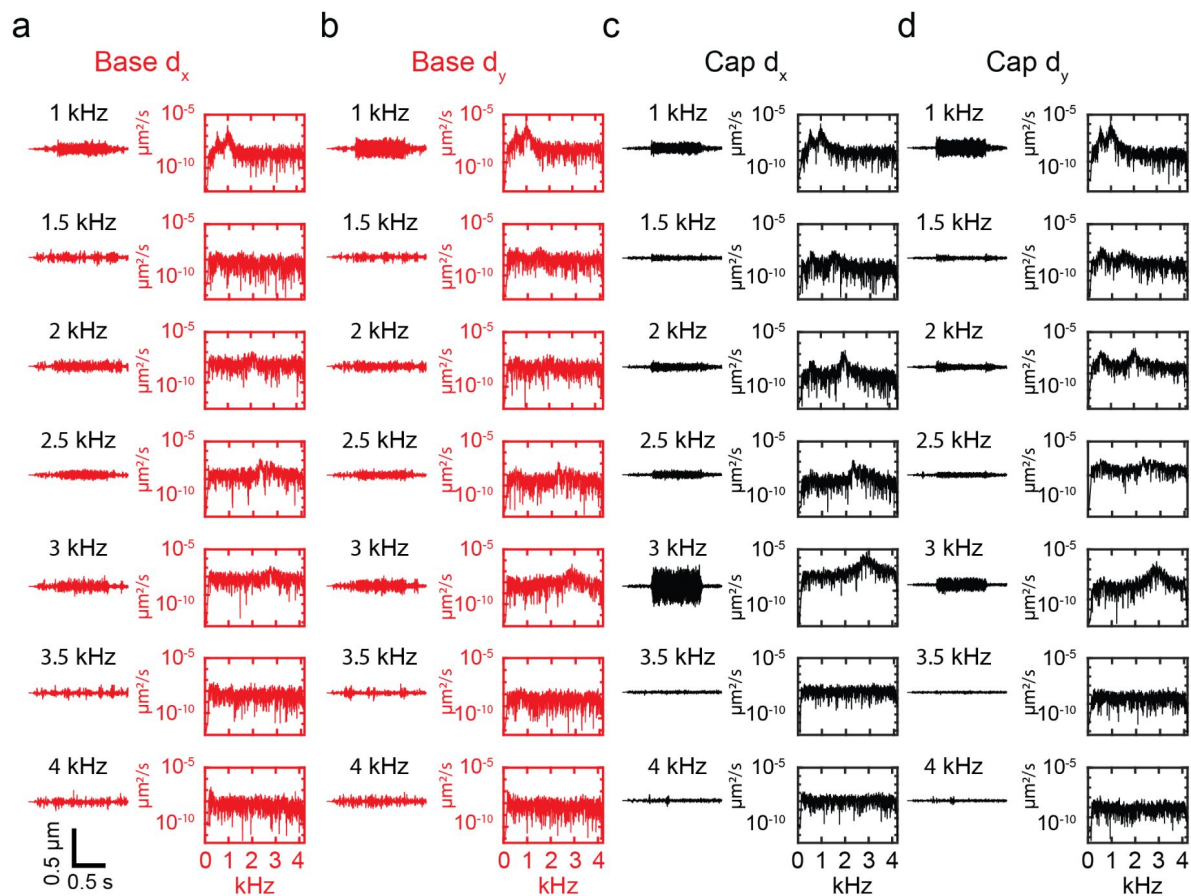
(a-c) Locations of the three groups of scolopidia in Müller's organ: Group-I scolopidia, located near the folded body; Group-II, located within the fusiform body, forming a 'bridge' to the pyriform vesicle at the surface of the tympanic membrane; and Group-III, located on the styliform body. The overlay indicates examples of scolopale sheaths (opaque yellow) and the corresponding attachment cells (blue). Dashed lines outline the dendrites along their lengths, and yellow arrowheads mark the other visible scolopidia. (d) Oblique section of the distal region of a scolopidium reveals the full extent of the cilium. (e) Example of a potential sample preparation artifact showing the apparent detachment of a cilium from its terminal region. (f) Longitudinal section through the ciliary dilation, revealing three central fibrils (arrowhead). (g) Longitudinal section along the dashed line in (f), reveals two of the central fibrils (arrowhead) extending along the entire length of the dilation.



Extended data figure 2.

Locations of Group-III scolopidia and frequency response of Müller's organ using OCT.

(a) Maximal intensity projection of a Müller's organ labelled with DAPI superimposed with manual registration of dendrite dilations (red circles). Anatomical landmarks are indicated at the maximal width of the ganglion (cyan dashed line) and the styliform body (yellow dashed line). (b) Maximal intensity projection of auditory neurons labelled with DyLight™ 488-conjugated streptavidin neurobiotin superimposed with manual measurement of dendrite dilations' positions (red circles) and schematic illustrating the computation of the relative distance of each dendrite dilation between the ganglion (cyan dashed line) and the styliform body (yellow dashed line). (c) Normalised positions of dendrite dilation from an example of one Müller's organ. The normalised positions of the dendrite dilations are plotted as a function of distance across the ganglion's width shown as cyan dashed line in (b). A black dashed line indicates the normalised position at 52% between the ganglion and the styliform body. (d) Distances between dendrite dilations and the ciliary rootlet of the corresponding scolopidium ($N = 19$, $n = 46$). (e, f) Amplitude (e) and phase (f) response of Müller's organ along the antero-posterior axis ($N = 4$, $E = 4$) measured using OCT, plotted as a function of relative distance between the ganglion and styliform body. (g, h) Amplitude (g) and phase (h) response of Müller's organ along the meso-lateral axis ($N = 7$, $E = 12$) measured using OCT, plotted as a function of relative distance between the ganglion and styliform body. The vertical dashed lines in (e-h) indicate the normalized position of the Group-III scolopidia. All vibration amplitudes are shown in dB relative to a 1-nm reference vibration. For additional data, see Source Data Extended Fig. 2 [2](#).



Extended data figure 3.

Example recordings of sound-evoked motion of a scolopidium.

Raw time series amplitude traces (left traces) and their corresponding Fourier transforms (right traces) for the scolopale base and cap along (dx) and orthogonal (dy) to the ciliary axis at each stimulated frequency from 1 to 4 kHz with an increment of 0.5 kHz. **(a, b)** Motion of the scolopale base (a) along and (b) orthogonal to the ciliary axis. **(c, d)** Motion of the cap (c) along and (d) orthogonal to the ciliary axis. Pure-tone sound stimulation at 90 dB SPL.

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Additional information

Author contributions

A.C. and B.W. designed and planned the experiments. A.C. performed sample preparation, acquisition, annotation, and three-dimensional reconstruction for Focus-Ion Beam Scanning Electron Microscopy (FIB-SEM). M.V.H. and M.N. performed the initial Optical Coherence Tomography (OCT) experiments, A.C. and B.W. finalised the data collection and analysis. B.W. performed whole-cell patch clamp recordings. A.C. and B.W. analysed the experimental data and prepared the initial version of the manuscript. M.N. and M.V.H. commented on it, and together with A.C. and B.W., finalised the submitted version of the manuscript.

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Additional files

Supplementary movie 1. High-resolution 3D ultrastructure of a scolopidium using Focused Ion Beam-Scanning Electron Microscopy (FIB-SEM). Example of the 3D ultrastructure of a scolopidium with annotations⁵⁶. [🔗](#)

Supplementary movie 2. High-speed imaging of the sound-evoked motion of the scolopidium. Each scolopidium motion is played at a different frame rate. The frame rates can be found with additional movies and their metadata is in a public depository Zenodo⁵⁶. This supplementary movie is based on the example Movie No. 1⁵⁵. [↗](#)

Supplementary movie 3. Mechanical stimulation of the scolopidium cap. The movie is played at 18.75 frames per second⁵⁶. [↗](#)

Source Data Figure 3 [↗](#)

Source Data Figure 4 [↗](#)

Source Data Figure 5 [↗](#)

Source Data Extended Data Figure 2 [↗](#)

Note

This reviewed preprint has been updated to add funding.

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Reviewer #1 (Public review):

Chaiyasitdhi et al. set out to investigate the detailed ultrastructure of the scolopidia in the locust Müller's organ, the geometry of the forces delivered to these scolopidia during natural stimulation, and the direction of forces that are most effective at eliciting transduction currents. To study the ultrastructure, they used the FIB-SEM technique, to study the geometry of natural stimulation, they used OCT vibrometry and high-speed light microscopy, and to study transduction currents, they used patch clamp physiology.

Strengths:

I believe that the ultrastructural description of the locust scolopidium is excellent and the first of its kind in any insect system. In particular, the finding of the bend in the dendritic cilium and the position of the ciliary dilation are interesting, and it would be interesting to see whether these are common features within the huge diversity of insect chordotonal organs.

I believe the use of OCT to measure organ movements is a significant strength of this paper; however, using ex vivo preparations undermines any conclusions drawn about the system's in vivo mechanics.

The choice of Group III scolopidia is also good. Research on the mechanics of locust tympana has shown that travelling waves are formed on the tympanum and waves of different frequencies show highest amplitudes at different positions on the tympanum, and therefore also on different groups of scolopidia within the Müller's organ (Windmill et al, 2005; 2008, and Malkin et al, 2013). The lowest frequency modal waves (F0) observed by Windmill et al 2008 were at about 4.4 kHz, which are slightly higher than the ~3 kHz frequencies studied in this paper but do show large deflections where these group III scolopidia attach at the styliform body (Windmill et al, 2005).

This should be mentioned in the paper since the electrophysiology justification to use group III neurons is less convincing, given that Jacobs et al 1999 clearly point out that group III neurons are very variable and some of them are tuned much higher to 10 kHz, and others even higher to 20-30 kHz.

Weaknesses:

Specifically, it is understandable that the authors decided to use excised ears for the light microscopy, where Müller's organ would not be accessible in situ. However, it is very likely that excision will change the system's mechanics, especially since any tension or support to Müller's organ will be ablated. OCT enables in vivo measurements in fully undissected systems (Mhatre et al, Biorxiv, 2021) or in systems with minimal dissection where the mechanics have not been compromised (Vavakou et al, 2021). The choice to entirely dissect out the membrane is difficult to understand here.

My main concern with this paper, however, is the use of light microscopy very close to the Nyquist limit to study scolopidial motion, and the fact that the OCT data contradict and do not match the light microscopy data.

The light microscopy data is collected at ~8 kHz, and hence the Nyquist limit is ~4 kHz. It is possible to measure frequencies reliably this close to the limit, but the amplitude of motion is quite likely to be underestimated, given that the technique only provides 2 sample points per cycle at 4 kHz and approximately 2.66 sample points at 3 kHz. At that temporal resolution, the samples are much more likely to miss the peak of the wave than not, and therefore, amplitudes will be misestimated. A much more reasonable sample rate for amplitude estimation is generally about 10 samples per cycle. I do not believe the data from the microscopy is reliable for what the authors wish to use them for.

Using the light microscopy data, the authors claim that the strains experienced by the group III scolopidia at 3 kHz are greater along the AP axis than the ML axis (Figure 4). However, this is contradicted by the OCT data, which show very low strain along the AP axis (black traces) at and around 3 kHz (Figure 3c and extended data Figure 2f) and show some movement along the ML axis (red traces, same figures). The phase at low amplitudes of motion cannot be considered very reliable either, and hence phase variations at these frequencies in the OCT cannot be considered reliable indicators of AP motion; hence, I'm unclear whether the vector difference in the OCT is a reliable indicator of movement.

The OCT data are significantly more reliable as they are acquired at an appropriate sampling rate of 90 kHz. The authors do not mention what microphone they use to monitor or calibrate their sound field and phase measurements in OCT, but I presume this was done since it is the norm. Thus, the OCT data show that the movement within the Müller's organ is complex, probably traces an ellipse at some frequencies as observed in bushcrickets (Vavkou et al, 2021) and also thought to be the case in tree crickets based on the known attachment points of the TO (Mhatre et al, 2021). The OCT data shows relatively low AP motion at frequencies near 3 kHz, and higher ML motion, which contradicts the less reliable light microscopy data. Given that the locust membrane shows peaks in motion at ~4.5 kHz, ~11 kHz, and also at ~20 kHz (Windmill et al, 2008), I am surprised that the authors limited their OCT experiments and analyses to 5 kHz.

In summary for this section, I am not convinced of the conclusion drawn by the authors that group III scolopidia receive significantly higher stimulation along the AP axis in their native configuration, if indeed they were studied in the appropriate force regime (altered due to excision).

In the scolopidial patch clamp data, the authors study transduction currents in response to steady state stimulation along the AP axis and the ML axis. The responses to steady state and

periodic forces may well be different, and the authors do not offer us a way to clearly relate the two and therefore, to interpret the data.

In addition, both stimulation types, along the AP axis and the ML, elicit clear transduction responses. Stimulation along the AP axis might be slightly higher, but there is over 40% variation around the mean in one case (pull: 26.22 {plus minus} 10.99 pA) and close to 80% variation in the other (push: 10.96 {plus minus} 8.59 pA). These data are indeed from a very high displacement range (2000 nm), which is very high compared to the native displacement levels, which are in the 1-10 nm range.

The factor change from sample to sample is not reported, and is small even overall. The statistical analyses of these data are not clearly reported, and I don't see the results of the overall ANOVA in the results section. I also find the dip in the reported transduction currents between 10 and 100 nm quite odd (Figure 5 j-m) and would like to know what the authors' interpretation of this behaviour is. It seems to me that those currents increase continuously linearly after ~50-100 nm and that the data below that range are in the noise. Thus, the transduction currents observed at the relevant displacement range (1-10 nm) may not actually be reliable. How were these small displacements achieved, and how closely were the actual levels monitored? Is it possible to reliably deliver 1-10 nm displacements using a micromanipulator?

What is clear, despite the difficulty in interpreting this data, is that both AP and ML stimulation evoke transduction currents, and their relative differences are small. Additionally, in Müller's organ itself, in the excised organ, the scolopidia are stimulated along both axes. Thus, in my opinion, it is not possible to say that axial stretch along the cilium is 'the key mechanical input that activates mechano-electrical transduction'.

<https://doi.org/10.7554/eLife.108740.1.sa3>

Reviewer #2 (Public review):

Summary of strengths and weaknesses:

Using several techniques-FIB-SEM, OCT, high-speed light microscopy, and electrophysiology-Chaiyasitdhi et al. provide evidence that chordotonal receptors in the locust ear (Müller's organ) sense the stretch of the scolapale cell, primarily of its cilium. Careful measurements certainly show cell stretch, albeit with some inconsistencies regarding best frequencies and amplitudes. The weakest argument concerns the electrophysiological recordings, because the authors do not show directly that the stimulus stretches the cells. If this latter point can be clarified, then our confidence that ciliary stretch is the proximal stimulus for mechanotransduction will be increased. This conclusion will not come as a surprise for workers in the field, as the chordotonal organ is known as a stretch-receptor organ (e.g., Wikipedia). But it is a useful contribution to the field and allows the authors to suggest transduction mechanisms whereby ciliary stretch is transduced into channel opening.

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Reviewer #3 (Public review):

Summary:

The paper 'A stretching mechanism evokes mechano-electrical transduction in auditory chordotonal neurons' by Chaiyasitdhi et al. presents a study that aims to address the mechanical model for scolopidia in *Schistocerca gregaria* Müller's organ, the basic mechanosensory units in insect chordotonal organs. The authors combine high-resolution

ultrastructural analysis (FIB-SEM), sound-evoked motion tracking (OCT and high-speed light microscopy), and electrophysiological recordings of transduction currents during direct mechanical stimulation of individual scolopidia. They conclude that axial stretching along the ciliary axis is an adequate mechanical stimulus for activating mechanotransduction channels.

Strengths/Highlights:

- (1) The 3D FIB-SEM reconstruction provides high resolution of scolopidial architecture, including the newly described "scolopale lid" and the full extent of the cilium.
- (2) High-speed microscopy clearly demonstrates axial stretch as the dominant motion component in the auditory receptors, which confirms a long-standing question of what the actual motion of a stretch receptor is upon auditory stimulation.
- (3) Patch-clamp recordings directly link mechanical stretch to transduction currents, a major advance over previous indirect models.

Weaknesses/Limitations:

- (1) The text is conceptually unclear or written in an unclear manner in some places, for example, when using the proposed model to explain the sensitivity of Nanchung-Inactive in the discussion.
- (2) The proposed mechanistic models (direct-stretch, stretch-compression, stretch-deformation, stretch-tilt) are compelling but remain speculative without direct molecular or biophysical validation. For example, examining whether the organ is pre-stretched and identifying the mechanical components of cells (tissues), such as the extracellular matrix and cytoskeleton, would help establish the mechanical model and strengthen the conclusion.
- (3) To some extent, the weaknesses of the paper are part of its strengths and vice versa. For example, the direct push/pull and up/down stimulations are a great experimental advance to approach an answer to the question of how the underlying cellular components are deformed and how the underlying ion channels are forced. However, as the authors clearly state, neither of their stimulations can limit all forces to only one direction, and both orthogonal forces evoke responses in the neurons. The question of which of the two orthogonal forces 'causes' the response cannot be answered with these experiments and has not been answered by this manuscript. But the study has brought the field a considerable step closer to answering the question. The answer, however, might be that both longitudinal ('stretch') and perpendicular ('compression') forces act together to open the ion channels and that both dendritic extension via stretch and bending can provide forces for ion channel gating. The current paper has identified major components (longitudinal stretch components) for the neurons they analysed, but these will surely have been chosen according to their accessibility, and as such, the variety of mechanical responses in Müller's organ might be greater. In light of these considerations, the authors might acknowledge such uncertainties more clearly in their paper. The paper is an impressive methodological progress and breakthrough, but it simply does not "demonstrate that axial stretch along the cilium is the adequate stimulus or the key mechanical input that activates mechano-electrical transduction" as the authors write at the start of their discussion. They do show that axial stretch dominates for the neurons they looked at, which is important information. The same applies to the end of the discussion: The authors write, "This relative motion within the organ then drives an axial stretch of the scolopidium, which in turn evokes the mechano-electrical transduction current." Reading the manuscript, the certainty and display of confidence are not substantiated by the data provided. But they are also not necessary. The study has paved the road to answer these questions. Instead, the authors are encouraged to make suggestions on how the remaining uncertainties could be removed (and what experiments or model might be used).

Author response:

Reviewer #1 (Public review):

Chaiyasitdhi et al. set out to investigate the detailed ultrastructure of the scolopidia in the locust Müller's organ, the geometry of the forces delivered to these scolopidia during natural stimulation, and the direction of forces that are most effective at eliciting transduction currents. To study the ultrastructure, they used the FIB-SEM technique, to study the geometry of natural stimulation, they used OCT vibrometry and high-speed light microscopy, and to study transduction currents, they used patch clamp physiology.

Strengths:

I believe that the ultrastructural description of the locust scolopidium is excellent and the first of its kind in any insect system. In particular, the finding of the bend in the dendritic cilium and the position of the ciliary dilation are interesting, and it would be interesting to see whether these are common features within the huge diversity of insect chordotonal organs.

Thank you very much for your comments. We indeed plan to extend and continue our approach to exploit and understand diverse chordotonal organs in insects and crustaceans.

I believe the use of OCT to measure organ movements is a significant strength of this paper; however, using ex vivo preparations undermines any conclusions drawn about the system's in vivo mechanics.

Having re-read the manuscript, we failed to explicitly describe our ex vivo preparation of Müller's organ including key references that detail the largely retained physiological function of Müller's organ. We have now revised this detail in the method section:

“We used an excised locust ear preparation for all experiments, following a previously described dissection protocol [9]. In short, the tympanum, with Muller's organ attached was left intact suspended between the cuticular rim. The cuticular rim of the tympanum was fixed into a hole in a preparation dish that allowed Muller's organ to be submerged with extracellular saline, whilst the outside of the tympanum was dry and could be stimulated with airborne sound. This ex vivo preparation of Muller's organ retained frequency tuning (Warren & Matheson, 2018), similar electrophysiological function as freshly dissected Muller's organs (Hill, 1983a, 1983b; Michelsen, 1968: frequency discrimination in the locust ear by means of four groups of receptor cells), and amplitude coding (Warren & Matheson, 2018). Since Müller's organ is backed by an air-filled trachea in vivo, the addition of saline solution in the ex vivo preparation decreased its displacements ~100 fold due to a dampening effect (Warren et al., 2020).”

And in the last section of the introduction:

“Here, we combined FIB-SEM to resolve the 3D ultrastructure of a scolopidium, OCT and high-speed microscopy to examine sound-evoked motion at both the organ and individual scolopidium levels, and direct mechanical stimulation of the scolopale cap, where the ciliary tip is anchored, whilst simultaneously recording transduction currents. Here, Muller's organ and the tympanum was excised from the locust for physiological experiments. This ex vivo preparation of Muller's organ retained frequency tuning, amplitude coding and electrophysiological function. This preparation also permitted the enzymatic isolation of individual scolopidia whilst recording transduction currents (Warren & Matheson, 2018).”

To further clarify physiological differences between the in vivo and ex vivo operation of the tympanum and Müller's organ, we will perform an additional experiment for the revised manuscript by quantifying the changes in the sound-evoked tonotopic travelling wave of the tympanum using Laser Doppler Vibrometry (LDV). This result will be added to the Supplementary Text.

The choice of Group III scolopidia is also good. Research on the mechanics of locust tympana has shown that travelling waves are formed on the tympanum and waves of different frequencies show highest amplitudes at different positions on the tympanum, and therefore also on different groups of scolopidia within the Müller's organ (Windmill et al, 2005; 2008, and Malkin et al, 2013). The lowest frequency modal waves (F0) observed by Windmill et al 2008 were at about 4.4 kHz, which are slightly higher than the ~3 kHz frequencies studied in this paper but do show large deflections where these group III scolopidia attach at the styliform body (Windmill et al, 2005).

Thank you very much. We accept that the frequencies studied in this manuscript were lower than the lowest modal wave observed by Windmill et al., 2008. Other authors, according to Jacobs et al. 1999, found broad tuning from 3.4-3.74 kHz (Michelson et al., 1971) and 2-3.5 kHz (Halex et al., 1988). We settled on tuning previously measured for Group-III neurons in the same kind of preparation as in this manuscript, which was broadly around 3 kHz (Warren & Matheson, 2018).

This should be mentioned in the paper since the electrophysiology justification to use group III neurons is less convincing, given that Jacobs et al 1999 clearly point out that group III neurons are very variable and some of them are tuned much higher to 10 kHz, and others even higher to 20-30 kHz.

Looking at Fig. 7 from Jacobs et al., 1999, we indeed see that the four Group-III neurons recorded in this study are broadly tuned to 3-4 kHz. Often these tuning curves have threshold dips at higher frequencies at least 20 dB higher. We settled on the most sensitive frequency that we previously measured, and which also overlaps the most sensitive frequencies from several other studies.

Weaknesses:

Specifically, it is understandable that the authors decided to use excised ears for the light microscopy, where Müller's organ would not be accessible in situ. However, it is very likely that excision will change the system's mechanics, especially since any tension or support to Müller's organ will be ablated.

We completely understand this criticism. We have now added descriptions in the methodology and introduction (as detailed previously). In short, the tympanum was left intact suspended on the cuticle. Müller's organ retains all (measured) physiological properties: frequency tuning, amplitude coding and electrophysiological function. To further investigate whether this excised preparation is a representative of the in vivo conditions, we plan to measure tympanal mechanics, such as the travelling wave, as part of the revisions.

OCT enables in vivo measurements in fully undissected systems (Mhatre et al, Biorxiv, 2021) or in systems with minimal dissection where the mechanics have not been compromised (Vavakou et al, 2021). The choice to entirely dissect out the membrane is difficult to understand here.

The pioneering OCT works by Mhatre et al, Biorxiv, 2021 and Vavakou et al, 2021 set the new standard of in vivo measurements in the field. We also totally agree with Reviewer#1's view

that OCT is best performed on in vivo Müller's organ and we tried OCT imaging of Müller's organ for several months in vivo. Although the OCT penetrates the tympanum the OCT beam does not penetrate the tracheal air sac that surrounds Müller's organ and therefore OCT cannot be used in vivo. Please also see previous comment with regards to the intact physiological operation of Muller's organ in the ex vivo preparation.

My main concern with this paper, however, is the use of light microscopy very close to the Nyquist limit to study scolopidial motion, and the fact that the OCT data contradict and do not match the light microscopy data. The light microscopy data is collected at ~8 kHz, and hence the Nyquist limit is ~4 kHz. It is possible to measure frequencies reliably this close to the limit, but the amplitude of motion is quite likely to be underestimated, given that the technique only provides 2 sample points per cycle at 4 kHz and approximately 2.66 sample points at 3 kHz. At that temporal resolution, the samples are much more likely to miss the peak of the wave than not, and therefore, amplitudes will be mis-estimated. A much more reasonable sample rate for amplitude estimation is generally about 10 samples per cycle. I do not believe the data from the microscopy is reliable for what the authors wish to use them for.

We understand your concern that the study of sound-evoked motion of the scolopidium using light microscopy was done near the Nyquist limit (with our average sampling rate at 8.6 ± 0.3 kHz and the Nyquist limit at 4.3 kHz). We also agree with your comment that amplitude of the motion could be underestimated at frequencies closer to the limit. However, we find that this systematic error does not change the key observation from our direct light microscopy observation that axial stretch of the scolopidium occurs around 3 kHz.

To address this concern, we plan to study the scolopidial motion within Group 1 auditory neurons, which are tuned to lower frequencies (0.5-1.5 kHz). This new set of data will allow us to obtain more data points per cycle (up to ~8.6 data points at 1 kHz). We will consider adding this result into the revised Fig. 4 or its extended data.

Regarding increasing the sampling rate, we did try to achieve higher sampling rate (> 10 kHz), however, there is a technical limitation of our camera and a trade-off between other key parameters, such as the size of the region of interest (ROI) and magnification. To increase the sampling rate, we will have to reduce the magnification or the ROI and in turn lose the spatial resolution required for quantification of the scolopidial motion or the ROI does not cover the whole scolopidial motion. The sampling rate at 8.6 ± 0.3 kHz was the best we could achieve.

Using the light microscopy data, the authors claim that the strains experienced by the group III scolopidia at 3 kHz are greater along the AP axis than the ML axis (Figure 4). However, this is contradicted by the OCT data, which show very low strain along the AP axis (black traces) at and around 3 kHz (Figure 3c and extended data Figure 2f) and show some movement along the ML axis (red traces, same figures). The phase at low amplitudes of motion cannot be considered very reliable either, and hence phase variations at these frequencies in the OCT cannot be considered reliable indicators of AP motion; hence, I'm unclear whether the vector difference in the OCT is a reliable indicator of movement.

This is our fault for not clearly explaining the orientation of the light microscopy measurement, which then leads to the reviewer's concern about contradiction between OCT and light microscopy. Our OCT measurements was done along the Antero-Posterior (AP) and Mesio-Lateral axes (ML), while the axial stretch of the scolopidium occurs along the Dorso-Ventral (DV) axis. We recognise that the anatomical references in this manuscript can be confusing, and we tried to show the orientation of the scolopidium relative to Müller's organ in Fig. 3b. To further clarify the orientation of our observations, we will add anatomical references in Fig. 4a and Fig. 5a. in the revised manuscript.

As stated in our result section (Line 165-167)

“Notably, we could not resolve the Group-III scolopidia along the ventro-dorsal axis—which runs parallel to the dendrite—as the OCT beam was obstructed by either the cuticle or the elevated process”

We did try to perform OCT measurement along the VD axis, but we could not resolve the scolopidial region along the scolopidial or ciliary axes because the OCT beam could not go through the thick cuticle at the edge of the tympanic membrane and the elevated process. For this reason, it is impossible for us to find an agreement or rule out any contradiction between the OCT and light microscopy since they are measuring motion along different axes. We plan to address this accessibility issue in a separate work using OCT measurements in combination with mirrors.

The OCT data are significantly more reliable as they are acquired at an appropriate sampling rate of 90 kHz. The authors do not mention what microphone they use to monitor or calibrate their sound field and phase measurements in OCT, but I presume this was done since it is the norm.

We use a condenser microphone (MK301, Microtech) and measuring amplifier (type 2610, Brüel & Kjær) for calibration. The calibration microphone was also calibrated beforehand using a sound calibrator type 4231 from B&K.

Thus, the OCT data show that the movement within the Müller's organ is complex, probably traces an ellipse at some frequencies as observed in bushcrickets (Vavkou et al, 2021) and also thought to be the case in tree crickets based on the known attachment points of the tympanal organ (Mhatre et al, 2021). The OCT data shows relatively low AP motion at frequencies near 3 kHz, and higher ML motion, which contradicts the less reliable light microscopy data. Given that the locust membrane shows peaks in motion at ~4.5 kHz, ~11 kHz, and also at ~20 kHz (Windmill et al, 2008), I am surprised that the authors limited their OCT experiments and analyses to 5 kHz.

We found that immediately above 5 kHz the displacements reduced to undetectable magnitudes. We accept that there may be other modes of vibration at higher frequencies >10 kHz (based on Jacobs et al., 1999) that we could have detected with OCT. However, we focused our analysis on Group-III neurons at the best frequency and frequencies that we could cross-compare between our high-speed imaging system and OCT.

In summary for this section, I am not convinced of the conclusion drawn by the authors that group III scolopidia receive significantly higher stimulation along the AP axis in their native configuration, if indeed they were studied in the appropriate force regime (altered due to excision).

Again, we accept our faults for not clearly displaying the anatomical references of the scolopidial and ciliary axes in Fig. 4 and Fig. 5. We also did not clearly describe in detail that our ex vivo preparation largely retains its physiological properties. We will address the errors of our measurement near Nyquist and provide additional information from Group 1 scolopidia where we could achieve higher data points per cycle.

In the scolopidial patch clamp data, the authors study transduction currents in response to steady state stimulation along the AP axis and the ML axis. The responses to steady state and periodic forces may well be different, and the authors do not offer us a way to clearly relate the two and therefore, to interpret the data.

We will revise the Fig. 5a to clarify that the push-pull were done along the Dorso-Ventral (DV) axis and the push-pull were done along the Antero-Posterior (AP) axis. We do agree that steady-state and periodic forces may well be very different. However, valuable insight can be gained from mechanical systems when displaced outside of their normal physiological frequency (e.g. the transformative work on vertebrate hair bundle mechanics, Howard & Hudspeth, 1988). For the same reason, we believe artificial stimulation of the scolopidium gives us new and crucial information to understand scolopodial mechanics. Our main finding that stretch is the dominant stimulus should still, or at least provide strong support, that stretch is the dominant stimulus in periodical motion.

In addition, both stimulation types, along the AP axis and the ML, elicit clear transduction responses. Stimulation along the AP axis might be slightly higher, but there is over 40% variation around the mean in one case (pull: 26.22 {plus minus} 10.99 pA) and close to 80% variation in the other (push: 10.96 {plus minus} 8.59 pA). These data are indeed from a very high displacement range (2000 nm), which is very high compared to the native displacement levels, which are in the 1-10 nm range.

In this experiment, we wished to establish the upper limits (and plateau region) of displacement-transduction current response. However, even at 2000 nm we still did not see a plateau. Therefore, we believe that the strain on the scolopidium is still in the operating range even though our displacement is not. This discrepancy can be explained because the base of the scolopidium is not fixed. Therefore, the displacement imposed in our experiment is not equivalent to the strain on the cilium but a combination of pulling and stretching along the length of the dendrite. The force, however, remains along that particular axis, supporting our main finding.

Another important consideration is that the cilium is surrounded by the scolopale wall. It is assumed that the scolopale wall is far stiffer than the ciliary and will therefore limit the amount of ciliary strain.

The factor change from sample to sample is not reported and is small even overall. The statistical analyses of these data are not clearly reported, and I don't see the results of the overall ANOVA in the results section.

We reported the statistical analyses in the Fig. 5 Source Data. We will now add tables displaying these statistics in the supplementary text of the revised manuscript.

I also find the dip in the reported transduction currents between 10 and 100 nm quite odd (Figure 5 j-m) and would like to know what the authors' interpretation of this behaviour is. It seems to me that those currents increase continuously linearly after ~50-100 nm and that the data below that range are in the noise. Thus, the transduction currents observed at the relevant displacement range (1-10 nm) may not actually be reliable. How were these small displacements achieved, and how closely were the actual levels monitored? Is it possible to reliably deliver 1-10 nm displacements using a micromanipulator?

One interpretation is that the cilium has both sensitive and insensitive mechanically gated ion channels. A finding that is also supported by Effertz et al., 2012. We will add a sentence in the discussion highlighting this interpretation. We will also provide our calibration of displacement vs voltage delivered to the piezo in the Supplementary Text.

What is clear, despite the difficulty in interpreting this data, is that both AP and ML stimulation evoke transduction currents, and their relative differences are small. Additionally, in Müller's organ itself, in the excised organ, the scolopidia are stimulated

along both axes. Thus, in my opinion, it is not possible to say that axial stretch along the cilium is 'the key mechanical input that activates mechano-electrical transduction'.

We confirm that the scolopidia are displaced along both. We also note that displacements of the scolopidium limited to the up-down axis will also produce a strain on the scolopidium along the push-pull axis. However, we tried to disentangle this complex motion by limiting the displacements to one axis during recordings of the transduction current. We found that displacement along the scolopodial axis generated the largest transduction currents. Even though there is large variation our statistical analysis confirmed a significant difference as stated in the result section (Line 283 – 286)

“Additionally, the transduction current evoked by pull from the resting position was larger than displacement upward, 12.17 ± 5.37 pA ($N = 11$, $n = 11$) (Tukey's procedure, $p = 1.75e-03$, $t = -3.83$) or downward 7.28 ± 9.76 pA ($N = 11$, $n = 11$) (Tukey's procedure, $p = 5.10e-06$, $t = -4.53$).”

The reason for large variation is that the discrete depolarisations (random depolarisations of unknown function and a common feature of chordotonal neurons so far recorded) have a similar magnitude to the transduction current produced by the step displacements. We will highlight these discrete depolarisations in Figure 4d and mention them in the results.

Reviewer #2 (Public review):

Summary of strengths and weaknesses:

Using several techniques-FIB-SEM, OCT, high-speed light microscopy, and electrophysiology-Chaiyasitdhi et al. provide evidence that chordotonal receptors in the locust ear (Müller's organ) sense the stretch of the scolopale cell, primarily of its cilium. Careful measurements certainly show cell stretch, albeit with some inconsistencies regarding best frequencies and amplitudes.

Thank you very much for acknowledging the strength of our study. Regarding the inconsistencies between best frequencies and amplitude, we believe that this concern largely arises from our faults for not clearly displaying the anatomical references of the scolopodial and ciliary axes in Fig. 4 and Fig. 5. As previously addressed in our response to Reviewer#1, we will add the anatomical references and revised the text to clarify the orientation of our measurements.

The weakest argument concerns the electrophysiological recordings, because the authors do not show directly that the stimulus stretches the cells. If this latter point can be clarified, then our confidence that ciliary stretch is the proximal stimulus for mechanotransduction will be increased.

We agree that the displacement is not solely stretching the scolopidium. However, the force is still constrained and acting along the push-pull axis. Due to this reason, we overestimate the displacement required to open the MET channels but stand by our conclusion that stretch is the dominant stimulus. For future work, we wish to devise a technique to mechanically clamp the base of the scolopidium and measure the more physiological relevant current-strain relationship.

This conclusion will not come as a surprise for workers in the field, as the chordotonal organ is known as a stretch-receptor organ (e.g., Wikipedia). But it is a useful contribution to the field and allows the authors to suggest transduction mechanisms whereby ciliary stretch is transduced into channel opening.

One of the goals of this manuscript is to highlight the lack of direct evidence for stretch-sensitivity of chordotonal organs, as this is assumed from their structure. More importantly the acceptance of chordotonal organs, as being stretch sensitive does not address the mechanism of how organs work. For instance, one candidate for the MET channel, NompC, is shown to be sensitive to compression (Wang et al., 2021). We find that a preconceived concept of “stretch-sensitive” mechanism, without an appreciation of scolopidium mechanics, cannot explain how NompC can be opened in chordotonal organs.

P. .E. Howse wrote in his work on ‘The Fine Structure and Functional Organisation of Chordotonal Organs’ in 1968 (Symp. Zool. Soc. Lon.) No. 23

“There is, however, a common tendency to refer to chordotonal organs in which scolopidia are contained in a connective tissue strand as “stretch receptor”. This is unfortunate in two senses, for firstly the implied function may not have been proved and secondly even if the organ responds to stretch the scolopidia may not.” then he proceeded to cite a pioneering work in the chordotonal organs of the hermit crab by R.C. Taylor (Comp. Biochem. Physiol. 1966) showing that the scolopidia may experience flexing when the connective strand are stretched.

This work represents the first efforts to investigate the problematic assumption of stretch-sensitivity of scolopidia since it was first highlighted 57 years ago.

Reviewer #3 (Public review):

Summary:

The paper 'A stretching mechanism evokes mechano-electrical transduction in auditory chordotonal neurons' by Chaiyasitdhi et al. presents a study that aims to address the mechanical model for scolopidia in Schistocerca gregaria Müller's organ, the basic mechanosensory units in insect chordotonal organs. The authors combine high-resolution ultrastructural analysis (FIB-SEM), sound-evoked motion tracking (OCT and high-speed light microscopy), and electrophysiological recordings of transduction currents during direct mechanical stimulation of individual scolopidia. They conclude that axial stretching along the ciliary axis is an adequate mechanical stimulus for activating mechanotransduction channels.

Strengths/Highlights:

- (1) The 3D FIB-SEM reconstruction provides high resolution of scolopidial architecture, including the newly described "scolopale lid" and the full extent of the cilium.*
- (2) High-speed microscopy clearly demonstrates axial stretch as the dominant motion component in the auditory receptors, which confirms a long-standing question of what the actual motion of a stretch receptor is upon auditory stimulation.*
- (3) Patch-clamp recordings directly link mechanical stretch to transduction currents, a major advance over previous indirect models.*

Weaknesses/Limitations:

- (1) The text is conceptually unclear or written in an unclear manner in some places, for example, when using the proposed model to explain the sensitivity of Nanchung-Inactive in the discussion.*

We will rephrase and make clearer the context of our findings for Nanchung-Inactive mechanism of MET in the introduction and the discussion. We will also refine and simplify unclear text overall.

(2) The proposed mechanistic models (direct-stretch, stretch-compression, stretch-deformation, stretch-tilt) are compelling but remain speculative without direct molecular or biophysical validation. For example, examining whether the organ is pre-stretched and identifying the mechanical components of cells (tissues), such as the extracellular matrix and cytoskeleton, would help establish the mechanical model and strengthen the conclusion.

We agree with the speculative nature of our four proposed hypotheses. We have, however, narrowed down from at least ten previous hypotheses (Field and Matheson, 1998). These hypotheses will enable us, and hopefully the field, to test them and more rapidly advance our understanding of how scolopidia work. We will add a section in the discussion as to the best way to experimentally test these four hypotheses (e.g pushing directly onto the cap should elicit sensitive responses for the cap-compression hypothesis).

(3) To some extent, the weaknesses of the paper are part of its strengths and vice versa. For example, the direct push/pull and up/down stimulations are a great experimental advance to approach an answer to the question of how the underlying cellular components are deformed and how the underlying ion channels are forced. However, as the authors clearly state, neither of their stimulations can limit all forces to only one direction, and both orthogonal forces evoke responses in the neurons. The question of which of the two orthogonal forces 'causes' the response cannot be answered with these experiments and has not been answered by this manuscript. But the study has brought the field a considerable step closer to answering the question. The answer, however, might be that both longitudinal ('stretch') and perpendicular ('compression') forces act together to open the ion channels and that both dendritic extension via stretch and bending can provide forces for ion channel gating.

Thank you very much for your acknowledgement of our experimental advances. We agree that this study cannot identify and localise the forces on the cilium as it is enclosed in the scolopidial unit. As previously explained, we plan to address this question in our next work by improving and expanding our experimental techniques, including modelling, to study the scolopidial mechanics based on our experiments using patch-clamp recording in combination with individual and direct manipulation the scolopidium.

The current paper has identified major components (longitudinal stretch components) for the neurons they analysed, but these will surely have been chosen according to their accessibility, and as such, the variety of mechanical responses in Müller's organ might be greater. In light of these considerations, the authors might acknowledge such uncertainties more clearly in their paper.

Our high-speed and OCT imaging confirms complex multi-dimensional displacements (and presumably forces) acting on the scolopidium. We agree that our mechanical stimulation cannot recapitulate such complex motions. But for future work we wish to extend our mechanical stimulation to three axis and also to pivot on the axis of the scolopidial cap.

The paper is an impressive methodological progress and breakthrough, but it simply does not "demonstrate that axial stretch along the cilium is the adequate stimulus or the key mechanical input that activates mechano-electrical transduction" as the authors write at the start of their discussion.

We rephrase to clarity that stretching along the “scolopidial axis”, not “along the ciliary axis” is the adequate stimulus. We cannot yet verify how this translates to forces acting on the

cilium, hence the four speculative hypotheses. We will re-write the discussion to make clear that we are only interpreting the forces and displacements at the level of the cilium.

They do show that axial stretch dominates for the neurons they looked at, which is important information. The same applies to the end of the discussion: The authors write, "This relative motion within the organ then drives an axial stretch of the scolopidium, which in turn evokes the mechano-electrical transduction current." Reading the manuscript, the certainty and display of confidence are not substantiated by the data provided. But they are also not necessary. The study has paved the road to answer these questions. Instead, the authors are encouraged to make suggestions on how the remaining uncertainties could be removed (and what experiments or model might be used).

We will moderate our conclusion in the discussion, but we are confident that we have experimental repeats, and the statistical test, to support our conclusion that stretching of the scolopidium provides that largest transduction current responses (although not at the level of the cilium). As mentioned previously, we will include a section in the discussion for the best way to test the hypotheses arising from this work.

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